CHAPTER 14

MODELS OF NETWORK FORMATION

A discussion of models of the formation of networks, particularly models of networks that grow by addition of vertices, such as the World Wide Web or citation networks

THE MODELS described in Chapters 12 and 13 provide an excellent tool for studying the structural features of networks, such as giant and small components, degree distributions, the lengths of paths in networks, and so forth. Moreover, as we will see in later chapters, they can also serve as a convenient basis for further modeling work, such as the modeling of network resilience or of the spread of diseases over contact networks.

But there is another important class of network model that has an entirely different purpose. In the models we have seen so far, the parameters of the network, such as the number of vertices and edges or the degree distribution, are fixed from the outset—chosen by the modeler to have some desired values. For instance, if we are interested in networks with power-law degree distributions, we make a random graph model with a power-law degree distribution as in Section 13.10 and then explore its structure analytically or computationally. But models of this kind offer no explanation of *why* the network should have a power-law degree distribution in the first place. In this chapter we describe models of a different kind that offer such an explanation.

The models in this chapter are *generative network models*. That is, they model the mechanisms by which networks are created. The idea behind models such as these is to explore hypothesized generative mechanisms to see what structures they produce. If the structures are similar to those of networks we observe in the real world, it suggests—though does not prove—that similar generative mechanisms may be at work in the real networks. The best-known example of a generative network model, and the one that we study first in this chapter, is the "preferential attachment" model for the growth of networks with power-law degree distributions. Later in the chapter we examine a number of other models, including generalizations of preferential attachment models, vertex copying models, and models based on optimization.

14.1 PREFERENTIAL ATTACHMENT

As discussed in Section 8.4, many networks are observed to have degree distributions that approximately follow power laws, at least in the tail of the distribution. Examples include the Internet, the World Wide Web, citation networks, and some social networks and biological networks. The power law is a somewhat unusual distribution and its occurrence in empirical data is often considered a potential indicator of interesting underlying processes.[203] A natural question to ask therefore is how might a network come to have such a distribution? This question was first directly considered in the 1970s by Price [275], who proposed a simple and elegant model of network formation that gives rise to power-law degree distributions.

See Section 4.2 for a discussion of citation networks.

Price was interested in, among other things, the citation networks of scientific papers, having authored an important early paper on the topic in the 1960s in which he pointed out the power-law degree distribution seen in these networks [274]. In considering the possible origins of the power law, Price was inspired by the work of economist Herbert Simon [299], who noted the occurrence of power laws in a variety of (non-network) economic data, such as the distribution of people's personal wealth. Simon proposed an explanation for the wealth distribution based on the idea that people who have money already gain more at a rate proportional to how much they already have. This seems a reasonable supposition. Wealthy individuals make money by investing the money they have, and the return on their investment is essentially proportional to the amount invested. Simon was able to show mathematically that this "rich-get-richer" effect can give rise to a power-law distribution and Price adapted Simon's methods, with relatively little change, to the network context. Price gave a name to Simon's mechanism:[204] he called it *cumulative advantage*, although it is more often known today by the name *preferential attachment*, which was coined in 1999 by Barabási and Albert [27]. In this book we use principally the latter term, which has become the accepted name in recent years.

Price's model of a citation network is as follows. We assume that papers are published continually (though they do not have to be published at a constant rate) and that newly appearing papers cite previously existing ones. As discussed in Section 4.2, the papers and citations form a directed citation network, the papers being the vertices and the citations being the directed edges between them. Since no paper ever disappears after it is published, vertices in this network are created but never destroyed.

Let the average number of papers cited by a newly appearing paper be c. In the language of graph theory, c is the average out-degree of the network; in the language of publishing, c is the average size of the bibliography of a paper. The model allows for the actual sizes of the bibliographies to fluctuate about c. So long as the distribution of sizes satisfies a few basic sanity conditions,[205] only the average value is important for the behavior of the model in the limit of large network size. In real citation networks the sizes of bibliographies also vary from one field to another and depend on when papers were published, the average bibliography having grown larger over the years in most fields, but these effects are neglected in the model.

The crucial central assumption of Price's model is that a newly appearing paper cites previous ones chosen at random with probability proportional to the *number of citations those previous papers already have*. In this most basic of models there is no question of which papers are most relevant topically or which papers are most original or best written or the difference between research articles and reviews, or any of the many other factors that certainly affect real citation patterns. The model is thus very much a simplified representation of the citation process. As we

have seen with the random graphs of previous chapters, however, even simple models can lead to real insights. We certainly need to remember that the model only represents one aspect of the citation process—and a hypothetical one at that—but with this in mind let us press on and see what we can discover.

As with personal wealth, it is not implausible that the number of citations a paper receives could increase with the number it already has. When one reads papers, one often looks up the other works that those papers cite and reads some of them too. If a work is cited often, then, all other things being equal, we are more likely to come across it than a less cited work. And if we read it and like it, then perhaps we will cite it ourselves if we write a paper on the same topic. This does not mean that the probability of a paper receiving a citation is precisely proportional to the number of citations the paper has already, but it does at least give some justification for why the rich should get richer in this paper citation context.

In fact, upon further thought, it's clear that the probability of receiving a new citation cannot be *precisely* proportional to the number of citations a paper already has. Except under unusual circumstances, papers start out life with zero citations, which, with a strict proportionality rule, would mean that their probability of getting new citations would also be zero and so they would have zero citations for ever afterwards. To get around this hitch, Price proposed that in fact the probability that a paper receives a new citation should be proportional to the number that it already has plus a positive constant a. (In fact, Price only considered one special case $a = 1$ in his original model, but there seems to be no particular reason to limit ourselves to this case, so we will treat the case of general $a > 0$.)

The constant a in effect gives each paper a number of "free" citations to get it started in the race—each paper acts as though it started off with a citations instead of none. An alternative interpretation is that a certain fraction of citations go to papers chosen uniformly at random without regard for how many citations they current have, while the rest go to papers chosen in proportion to current citation count.[206] This gives all papers a chance to accrue citations, even if they currently have none. (We discuss this interpretation in more detail in Section 14.1.1, where we use it to construct a fast algorithm for simulating Price's model.)

We also need to specify what the starting state of the network is, how we initialize the model to begin with. It turns out in fact that in the limit of large network size the predictions of the model don't depend on the initial conditions, but we could, for instance, start the network out with a small set of initial papers having zero citations each.

Thus, in summary, Price's model consists of a growing network of papers and their citations in which vertices (papers) are continually added but none are ever taken away, each paper cites on average c others (so that the mean out-degree is c), and the cited papers are chosen at random[207] with probability proportional to their in-degree plus a constant a.

See Section 6.4.2 for a discussion of acyclic networks.

One important property of Price's model is immediately apparent: it generates purely acyclic networks, since every edge points from a more recently added vertex to a less recently added one, i.e., backward in time. Thus all directed paths in the network point backward in time and hence there can be no closed loops, because to close a loop we would need edges pointing forward in time as well. This fits well with the original goal of the model as a model of citation, since citation networks are acyclic, or very nearly so (see Section 4.2). On the other hand it fits poorly with some other directed networks such as the World Wide Web, although the model is still sometimes used as a model for power-law distributions in the Web.

Armed with our definition of Price's model, we will now write down equations governing the distribution of the in-degrees of vertices, i.e., the numbers of citations received by papers in terms of the parameters c and a, and hence solve for the degree distribution and various other quantities, at least in the limit of large network size. We will discuss models of both directed and undirected graphs in this chapter, so we will need to be careful to distinguish in-degree in the directed case from ordinary undirected degree in the undirected case. Previously in this book we have done this by denoting the in-degree of a vertex i by k_i^{in} (see Section 6.9), but this notation can make our equations quite difficult to read, so in the interests of clarity we will in this chapter adopt instead

the notation introduced by Dorogovtsev *et al.* [99] in which the in-degree of vertex i is denoted q_i. Degrees in undirected graphs will still be denoted k_i just as before.

So consider Price's model of a growing network and let $p_q(n)$ be the fraction of vertices in the network that have in-degree q when the network contains n vertices—this is the in-degree distribution of the network—and let us examine what happens when we add a single new vertex to the network.

Consider one of the citations made by this new vertex. In the model, the probability that the citation is to a particular other vertex i is proportional to $q_i + a$, where a is a positive constant. Since the citation in question has to be to *some* paper, this probability must be normalized such that its sum over all i is 1. In other words the correctly normalized probability must be

$$\frac{q_i + a}{\sum_i (q_i + a)} = \frac{q_i + a}{n\langle q \rangle + na} = \frac{q_i + a}{n(c + a)},$$

(14.1)

where we have written the average in-degree as $\langle q \rangle = n^{-1} \sum_i q_i$. In the second equality we have made use of the fact that the average out-degree of the network is c by definition, and that the average in- and out-degrees of a directed network are equal (see Eq. (6.27)) so that $\langle q \rangle = c$.

Each newly appearing paper cites c others on average, so the expected number of new citations to vertex i upon appearance of our new paper is c times Eq. (14.1). And there are $np_q(n)$ vertices with degree q in our network and hence the expected number of new citations to all vertices with degree q is

$$np_q(n) \times c \times \frac{q + a}{n(c + a)} = \frac{c(q + a)}{c + a} p_q(n).$$

(14.2)

Now we can write down a so-called *master equation* for the evolution of the in-degree distribution as follows. When we add a single new vertex to our network of n vertices, the number of vertices in the network with in-degree q increases by one for every vertex previously of in-degree $q - 1$ that receives a new citation,[208] thereby becoming a vertex of in-degree q. From Eq. (14.2) we know that the expected number of such vertices is

$$\frac{c(q - 1 + a)}{c + a} p_{q-1}(n).$$

(14.3)

Similarly, we lose one vertex of in-degree q every time such a vertex receives a new citation, thereby becoming a vertex of in-degree $q + 1$. The expected number of such vertices receiving citations is

$$\frac{c(q+a)}{c+a}p_q(n).$$

(14.4)

The number of vertices with in-degree q in the network after the addition of a single new vertex is $(n+1)p_q(n+1)$ which, putting together the results above, is given by

$$(n+1)p_q(n+1) = np_q(n) + \frac{c(q-1+a)}{c+a}p_{q-1}(n) - \frac{c(q+a)}{c+a}p_q(n).$$

(14.5)

The first term on the right-hand side here represents the number of vertices previously of in-degree q, the second term represents the vertices gained, and the third term the vertices lost.

Equation (14.5) applies for all values of q except $q = 0$. When $q = 0$ there are no vertices of lower degree that can gain an edge to become vertices of degree zero, and hence the second term in Eq. (14.5) doesn't appear. On the other hand, we gain a vertex of degree zero whenever a new vertex is added to the network, since by hypothesis papers have no citations when they are first published. Since exactly one vertex is added in going from a network of n vertices to a network of $n+1$, the appropriate equation for $q = 0$ is:

$$(n+1)p_0(n+1) = np_0(n) + 1 - \frac{ca}{c+a}p_0(n).$$

(14.6)

Now let us consider the limit of large network size $n \to \infty$ and calculate the asymptotic form of the degree distribution in this limit.[209] Taking the limit $n \to \infty$ and using the shorthand $p_q = p_q(\infty)$, Eqs. (14.5) and (14.6) become

$$p_q = \frac{c}{c+a}[(q-1+a)p_{q-1} - (q+a)p_q] \qquad \text{for } q \geq 1,$$

(14.7)

$$p_0 = 1 - \frac{ca}{c+a}p_0 \qquad \text{for } q = 0.$$

(14.8)

The second of these equations we can easily rearrange to give an explicit expression for the fraction p_0 of degree-zero vertices:

$$p_0 = \frac{1 + a/c}{a + 1 + a/c}.$$

(14.9)

The solution for $q \geq 1$ is a little more complicated, though only a little. Rearranging Eq. (14.7) for p_q we find that

$$p_q = \frac{q + a - 1}{q + a + 1 + a/c} p_{q-1}.$$

(14.10)

We can use this equation to calculate p_q iteratively for all values of q starting from our solution for p_0, Eq. (14.9). First, we set $q = 1$ in Eq. (14.10) to get

$$p_1 = \frac{a}{a + 2 + a/c} p_0 = \frac{a}{(a + 2 + a/c)} \frac{(1 + a/c)}{(a + 1 + a/c)}.$$

(14.11)

Now we can use this result to calculate p_2:

$$p_2 = \frac{a + 1}{a + 3 + a/c} p_1 = \frac{(a + 1)a}{(a + 3 + a/c)(a + 2 + a/c)} \frac{(1 + a/c)}{(a + 1 + a/c)},$$

(14.12)

and

$$p_3 = \frac{a+2}{a+4+a/c}\,p_2 = \frac{(a+2)(a+1)a}{(a+4+a/c)(a+3+a/c)(a+2+a/c)}\,\frac{(1+a/c)}{(a+1+a/c)},$$

(14.13)

and so forth. It's easy to see that for general q the correct expression must be

$$p_q = \frac{(q+a-1)(q+a-2)\ldots a}{(q+a+1+a/c)\ldots(a+2+a/c)}\,\frac{(1+a/c)}{(a+1+a/c)}.$$

(14.14)

This is effectively a complete solution for the degree distribution of Price's model, but there is a little more we can do to write it in a useful form. We make use of the gamma function,

$$\Gamma(x) = \int_0^\infty t^{x-1}\mathrm{e}^{-t}\,\mathrm{d}t,$$

(14.15)

which has the useful property that

$$\Gamma(x+1) = x\Gamma(x)$$

(14.16)

for all $x > 0$. Iterating this formula, we see that

$$\frac{\Gamma(x+n)}{\Gamma(x)} = (x+n-1)(x+n-2)\ldots x.$$

(14.17)

Using this result in Eq. (14.14) we can write

$$p_q = (1 + a/c)\frac{\Gamma(q+a)\Gamma(a+1+a/c)}{\Gamma(a)\Gamma(q+a+2+a/c)}.$$

(14.18)

This expression can be simplified further by writing it in terms of Euler's beta function, which is defined by

$$\mathrm{B}(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}.$$

(14.19)

If we multiply both the numerator and the denominator of Eq. (14.18) by $\Gamma(2 + a/c) = (1 + a/c)\Gamma(1 + a/c)$, we find that

$$p_q = \frac{\Gamma(q+a)\Gamma(2+a/c)}{\Gamma(q+a+2+a/c)} \times \frac{\Gamma(a+1+a/c)}{\Gamma(a)\Gamma(1+a/c)},$$

(14.20)

or

$$p_q = \frac{\mathrm{B}(q+a, 2+a/c)}{\mathrm{B}(a, 1+a/c)}.$$

(14.21)

Note that this expression is not only correct for $q \geq 1$ but also gives the correct value when $q = 0$.

One of the nice things about Eq. (14.21) is that it depends on q only via the first argument of the upper beta function. Thus if we want to understand the shape of the degree distribution we need only to understand the behavior of this one function.

In particular, let us examine the behavior for large q and fixed a and c. For large values of its first argument, we can rewrite the beta function using Stirling's approximation for the gamma function [2]

$$\Gamma(x) \simeq \sqrt{2\pi}\, \mathrm{e}^{-x} x^{x-\frac{1}{2}},$$

(14.22)

which means that

$$\mathrm{B}(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)} \simeq \frac{\mathrm{e}^{-x}x^{x-\frac{1}{2}}}{\mathrm{e}^{-(x+y)}(x+y)^{x+y-\frac{1}{2}}}\,\Gamma(y).$$

(14.23)

But

$$(x+y)^{x+y-\frac{1}{2}} = x^{x+y-\frac{1}{2}}\left[1+\frac{y}{x}\right]^{x+y-\frac{1}{2}} \simeq x^{x+y-\frac{1}{2}}\mathrm{e}^{y},$$

(14.24)

where the last equality becomes exact in the limit of large x. Then

$$\mathrm{B}(x,y) \simeq \frac{\mathrm{e}^{-x}x^{x-\frac{1}{2}}}{\mathrm{e}^{-(x+y)}x^{x+y-\frac{1}{2}}\,\mathrm{e}^{y}}\Gamma(y) = x^{-y}\,\Gamma(y).$$

(14.25)

In other words, the beta function B(x, y) falls off as a power law for large values of x, with exponent y.

Applying this finding to Eq. (14.21) we then discover that for large values of q the degree distribution of our network goes as $p_q \sim (q + a)^{-\alpha}$, or simply

$$p_q \sim q^{-\alpha}$$

(14.26)

when $q \gg a$, where the exponent α is

$$\alpha = 2 + \frac{a}{c}.$$

(14.27)

Thus Price's model for a citation network gives rise to a degree distribution with a power-law tail. This is very much in keeping with the degree distributions of real citation networks, which, as we saw in Fig. 8.8, appear to have clear power-law tails.

Note that the exponent $\alpha = 2 + a/c$ is strictly greater than two (since a and c are both strictly positive). Most measurements put the exponent of the power law for citation networks around $\alpha = 3$ (see Table 8.1), which is easily achieved in the model by setting the constants a and c equal. In a typical experimental situation the exponent α and the parameter c, the mean size of a paper's bibliography, are easily measured, but the parameter a, which represents the number of "free" effective citations a paper receives upon publication, is not. Typically therefore the value of a is extracted by rearranging Eq. (14.27) to give $a = c(\alpha - 2)$.

While it is delightful that Price's simple model generates a power-law degree distribution similar to that seen in real networks, we should not take the details of the model too seriously, nor the exact relation between the parameters and the exponent of the power law. As we noted at the start of this section, the model is highly simplified and substantially incomplete as a model of the citation process, omitting many factors that are undoubtedly important for real citations, including the quality and relevance of papers, developments and fashions in the field of study, the reputation of the publishing journal and of the author, and many others besides. Still, Price's model is striking in its ability to reproduce one of the most interesting features of citation networks using only a small number of reasonable assumptions, and many scholars believe that it may capture the fundamental mechanism behind the observed power-law degree distribution.

14.1.1 COMPUTER SIMULATION OF PRICE'S MODEL

When Price proposed his model in the 1970s, analytic treatments like the one above were essentially the only tool available for understanding the behavior of such models. Today, however, we can go further and study the operation of the model explicitly by performing computer simulations following the rules Price laid down. In addition to providing a useful check on our solution for the degree distribution, such simulations also allow us to generate real examples of networks on our computer. We can then measure these networks to determine the values, within the model, of any network quantities we like—path lengths, correlations, clustering coefficients, and so forth—including ones for which we do not at present have an analytic solution. Researchers have also made use of simulated networks as a convenient but still relatively realistic substrate for other kinds of calculation, including solutions of dynamical models, percolation processes, opinion formation models, and others.

In principle, simulation of Price's model appears straightforward. Typically one simulates the model with the out-degrees of vertices fixed to be exactly equal to c, where c is restricted to integer values. (In the original model and our analysis above, c was only the average out-degree—actually out-degree could fluctuate about the average.) Then the only complicated part of the simulation is the selection of the vertices that receive new edges, which has to be done in a random but non-uniform way as a function of the vertices' current in-degree. There are standard techniques for simulating such non-uniform random processes and one can without too much labor create a simple program that carry out the steps of the model. This, however, is not usually the best way to proceed. A naive direct simulation of this kind becomes slow when the network gets large, which limits the size of the network that can be generated. Luckily, there is a much faster way to perform the simulation that allows larger networks to be generated in shorter times, while still being simple to program on a computer. This method, first proposed by Krapivsky and Redner [187], works as follows.

When we create a new edge in Price's model we attach it to a vertex chosen in proportion to in-degree plus a constant a. Let us denote by θ_i the probability that an edge attaches to vertex i, which from Eq. (14.1) is given by

$$\theta_i = \frac{q_i + a}{n(c+a)}.$$

(14.28)

Now consider an alternative process in which upon creating a new edge we do one of two things. With some probability φ we attach the edge to a vertex chosen strictly in proportion to its current in-degree, i.e., with probability

$$\frac{q_i}{\sum_j q_j} = \frac{q_i}{nc}.$$

(14.29)

Alternatively, with probability $1 - \varphi$, we attach to a vertex chosen uniformly at random from all n possibilities, i.e., with probability $1/n$. Then the total probability θ'_i of attaching to vertex i in this process is

$$\theta'_i = \phi\frac{q_i}{nc} + (1-\phi)\frac{1}{n}.$$

(14.30)

Now let us make the choice $\varphi = c/(c + a)$, so that

$$\theta'_i = \frac{c}{c+a}\frac{q_i}{nc} + \left(1 - \frac{c}{c+a}\right)\frac{1}{n} = \frac{q_i + a}{n(c+a)}.$$

(14.31)

This, however, is precisely equal to the probability θ_i, Eq. (14.28), of selecting a vertex in the Price model and the two processes thus choose vertices with the exact same probabilities.

So an alternative way of performing a step of Price's model is the following:

> With probability $c/(c + a)$ choose a vertex in strict proportion to in-degree. Otherwise choose a vertex uniformly at random from the set of all vertices.

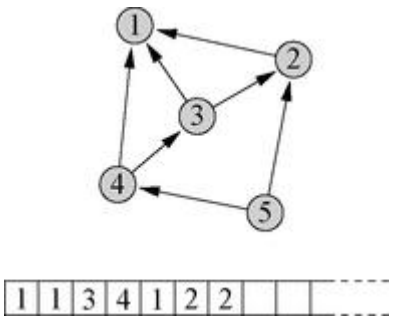


Figure 14.1: The vertex label list used in the simulation of Price's model. The list (bottom) contains one entry for the target of each edge in the network (top). In this example, there are three edges that point to vertex 1 and hence there are three elements containing the number 1 in the list. Similarly there are two containing the number 2, because vertex 2 is the target of two edges. And so forth.

The choice between the two parts can be achieved, for example, by generating a random number r in the range $0 \le r < 1$. If $r < c/(c + a)$ then we choose a vertex in proportion to in-degree. Otherwise we choose uniformly.

Choosing a vertex uniformly is easily accomplished. Choosing a vertex in proportion to in-degree is only slightly harder. It can be done rapidly by noting that choosing in proportion to in-degree is equivalent to picking an edge in the network uniformly at random and choosing the vertex which that edge points to. By definition this makes a vertex with in-degree q exactly q times as likely to be chosen as a vertex with in-degree 1, since it has q opportunities to be chosen, one for each of the edges that point to it.

To turn this observation into a computer algorithm we make a list, stored for instance in an ordinary array, of the target of each directed edge in the network. That is, the list's elements contain the vertex labels i of the vertices to which each edge points. Figure 14.1 shows an example for a small network. Note that the edges do not have to be in any particular order. Any order will do. Nor does the size of the array used to store the list have to match the length of the list exactly; it can contain empty elements at the end as shown in the figure. Indeed, since making already existing arrays larger is difficult in most computer languages, it makes sense to initially create an array that is large enough to hold the longest list we will need. (This means that it should have length nc if the out-degree of vertices is constant. If out-degree is allowed to fluctuate then the longest list might be a bit larger or smaller than nc, in which case one might create an array of size nc plus a few percent, to be on the safe side.)

Once we have our list, choosing a vertex in proportion to its in-degree becomes a trivial operation: we simply choose an element uniformly at random from the list and our vertex is identified by the contents of that element. When a new edge is added to the network, we must also update the list by adding the target of that edge to the end of the list.

Thus our algorithm for creating a new edge is the following:

1. Generate a random number r in the range $0 \le r < 1$.
2. If $r < c/(c + a)$, choose an element uniformly at random from the list of targets.
3. Otherwise choose a vertex uniformly at random from the set of all vertices.
4. Create an edge linking to the vertex thus selected, and add that vertex to the end of the list of targets. Each step in this process can be accomplished in constant time and hence the growth of a network of n vertices can be accomplished in time $O(n)$ (provided other parts of the program are implemented efficiently so that they also take constant time per step).

Figure 14.2a shows the degree distribution of a 100-million-node network generated computationally in this fashion, and the power-law form in the tail of the distribution is clearly visible. A practical problem, however, is the noise in the tail of the histogram, which makes the exact form of the distribution hard to gauge. This is exactly the same problem as we encountered for real-world data in Section 8.4.1: the bins in the tail of the histogram have relatively few samples in them and so the statistical fluctuations are large as a fraction of the number of samples. Indeed, in many respects simulation data often behave in similar ways to experimental data and they can often be treated using the same techniques. In this case we can take a hint from Section 8.4.1 and plot a cumulative distribution function instead of a histogram. To recap, the cumulative distribution function P_q is

$$P_q = \sum_{q'=q}^{\infty} p_{q'},$$

(14.32)

(see Eq. (8.4)) and is expected to have a power-law tail with an exponent $\alpha - 1 = 1 + a/c$, one less than the exponent of the degree distribution itself. Figure 14.2b shows the cumulative distribution of degrees for our simulation and we now see a much cleaner power-law behavior over several decades in q.

For comparison we can also calculate the cumulative distribution function analytically from our solution of the model. To do this, we make use of the standard integral form for the beta function:

$$\mathrm{B}(x,y) = \int_0^1 u^{x-1}(1-u)^{y-1}\,\mathrm{d}u.$$

(14.33)

Using this expression we find that

$$\begin{aligned} P_q &= \sum_{q'=q}^{\infty} p_q = \frac{1}{\mathrm{B}(a,1+a/c)} \sum_{q'=q}^{\infty} \int_0^1 u^{q'+a-1}(1-u)^{1+a/c}\,\mathrm{d}u \\ &= \frac{1}{\mathrm{B}(a,1+a/c)} \int_0^1 u^{a-1} \sum_{q'=q}^{\infty} u^{q'}(1-u)^{1+a/c}\,\mathrm{d}u \\ &= \frac{1}{\mathrm{B}(a,1+a/c)} \int_0^1 u^{q+a-1}(1-u)^{a/c}\,\mathrm{d}u \\ &= \frac{\mathrm{B}(q+a,1+a/c)}{\mathrm{B}(a,1+a/c)}. \end{aligned}$$

(14.34)

Given that $\mathrm{B}(x, y)$ goes as x^{-y} for large x (Eq. (14.25)), this implies that indeed the cumulative distribution function has a power-law tail with exponent $1 + a/c$.

In Fig. 14.2b we show Eq. (14.34) along with the simulation data, and the simulation and analytic solution agree well, as we would hope.

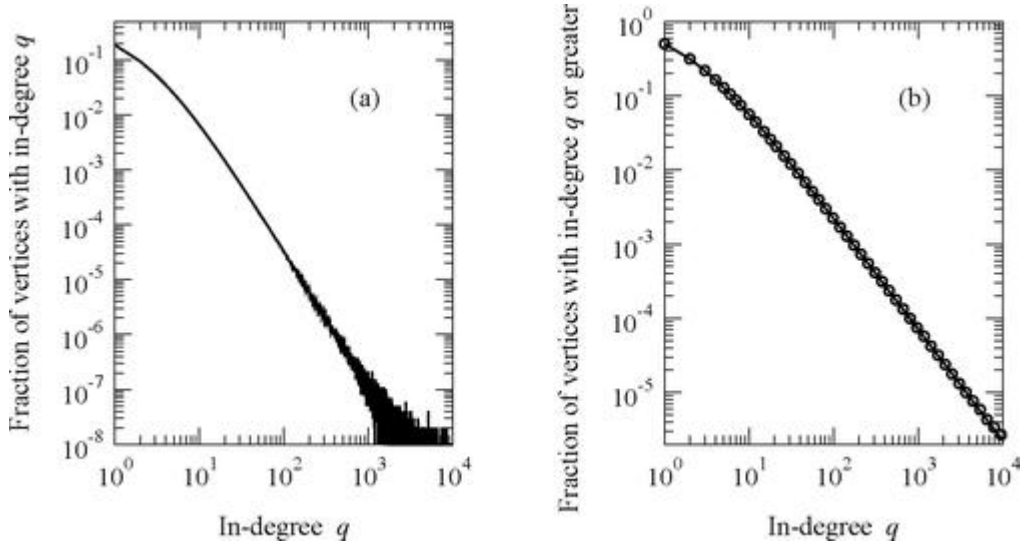


Figure 14.2: Degree distribution in Price's model of a growing network. (a) A histogram of the in-degree distribution for a computer-generated network with $c = 3$ and $a = 1.5$ which was grown until it had $n = 10^8$ vertices. The simulation took about 80 seconds on the author's computer using the fast algorithm described in the text. (b) The cumulative distribution function for the same network. The points are the results from the simulation and the solid line is the analytic solution, Eq. (14.34).

14.2 THE MODEL OF BARABÁSI AND ALBERT

Price's model of a growing network is an elegant one and the existence of an exact solution showing that its degree distribution has a power-law tail makes a persuasive case for preferential attachment as a possible origin for power-law behavior. At least until recently, however, Price's work in this area was not well known outside of the information science community. Preferential attachment did not become widely accepted as a mechanism for generating power laws in networks until much later, in the 1990s, when it was independently discovered by Barabási and Albert [27], who proposed their own model of a growing network (along with the name "preferential attachment"). The Barabási-Albert model, which is certainly the best known generative network model in use today, is similar to Price's, though not identical, being a model of an undirected rather than a directed network.

In the model of Barabási and Albert, vertices are again added one by one to a growing network and each vertex connects to a suitably chosen set of previously existing vertices. The connections, however, are now undirected and the number of connections made by each vertex is exactly c (unlike Price's model, where the number of connections was required only to take an average value of c but might vary from step to step). Note that this implies that c must be an integer, since a vertex cannot have non-integer degree. Connections are made to vertices with probability precisely proportional to the vertices' current degree. Notice that there is no in- or out-degree now because the network is undirected. Connections are made simply in proportion to the (undirected) degree. We will denote the degree of vertex i by k_i to distinguish it from the directed in-degree q_i of the last section. As before, vertices and edges are only ever added to the network and never taken way, which means, among other things, that there are no vertices with degree $k < c$. The smallest degree in the network is always $k = c$.

One can write down a solution for the model of Barabási and Albert using a master equation method similar to that of Section 14.1,[212] but in fact there is no need, because it is straightforward to show that the model is equivalent to a special case of Price's model. Imagine that, purely for the purposes of our discussion, we give each edge added to the network a direction, running from the vertex just added to the previously existing vertex that the edge connects to. That is each edge runs from the more recent of the two vertices it connects to the less recent. In this way we convert our network into a directed network in which each vertex has out-degree exactly c (since this is the number of outgoing edges a vertex starts with and it never gains any more). And the total degree k_i of a vertex in the sense of the original undirected network is the sum of the vertex's in-degree and out-degree, which is $k_i = q_i + c$ where q_i is the in-degree as before.

But given that the probability of an edge attaching to a vertex is simply proportional to k_i, it is thus also proportional to $q_i + c$, which is the same as in Price's model if we make the particular choice $a = c$. Thus the distribution of in-degrees in this directed network is the same as for Price's model with $a = c$, which we find from Eq. (14.21) to be

$$p_q = \frac{\mathrm{B}(q+c,3)}{\mathrm{B}(c,2)}. \tag{14.35}$$

To get the distribution of the total degree we then simply replace $q + c$ by k to get

$$p_k = \begin{cases} \dfrac{\mathrm{B}(k,3)}{\mathrm{B}(c,2)} & \text{for } k \geq c, \\ 0 & \text{for } k < c. \end{cases}$$

(14.36)

This expression can be simplified further by making use of Eq. (14.17) to write

$$\mathrm{B}(k,3) = \frac{\Gamma(k)\Gamma(3)}{\Gamma(k+3)} = \frac{\Gamma(3)}{k(k+1)(k+2)},$$

(14.37)

and similarly

$$\mathrm{B}(c,2) = \frac{\Gamma(2)}{c(c+1)},$$

(14.38)

so that

$$p_k = \frac{\Gamma(3)}{\Gamma(2)} \frac{c(c+1)}{k(k+1)(k+2)} = \frac{2c(c+1)}{k(k+1)(k+2)}$$

(14.39)

for $k \geq c$, where we have used (14.17) again to get rid of the remaining gamma functions. In the limit where k becomes large, this gives

$$p_k \sim k^{-3},$$

(14.40)

and hence the Barabási-Albert model generates a degree distribution with a power-law tail that

always has an exponent $\alpha = 3$.

Equation (14.39) was first derived by Krapivsky *et al.* [189] and independently by Dorogovtsev *et al.* [99]. A more detailed treatment was later given by Bollobás *et al.* [48], which clarifies precisely the domain of validity of the solution and the possible deviations from the expected value of p_k.

The model of Barabási and Albert can be simulated efficiently on a computer by exploiting the same mapping to Price's model and the simulation method described in Section 14.1.1. Again we regard the network as a directed one and maintain a list of targets of every directed edge, i.e., the vertices that the edges point to. Then, setting $a = c$, the algorithm of Section 14.1.1 becomes particularly simple: with probability $\frac{1}{2}$ we choose an element from our list uniformly at random and take the contents of that element as our target vertex. Otherwise we choose a target uniformly at random from the set of all vertices currently in existence. Then we create a new edge from the vertex just added to the target we have selected and also add that target to the end of our list.

The Barabási-Albert model is attractive for its simplicity—it doesn't require the offset parameter a of Price's model and hence has one less parameter to worry about. It is also satisfying that one can write the degree distribution without using special functions such as the beta and gamma functions that appear in the solution of Price's model. The price one pays for this simplicity is that the model can no longer match the exponents observed in real networks, being restricted to just a single exponent value $\alpha = 3$.

14.3 FURTHER PROPERTIES OF PREFERENTIAL ATTACHMENT MODELS

The models of Price and of Barabási and Albert were proposed as explanations for the observed power-law degree distributions in networks such as the Web, citation networks, and others, and for this reason the degree distribution is the property of these models that has attracted most interest. However, it is not the only property that can be calculated. The master equation method can be extended to the calculation of a number of other properties, many of which are interesting in their own right. We describe a few of these calculations in this section.

14.3.1 DEGREE DISTRIBUTION AS A FUNCTION OF TIME OF CREATION

Consider a network grown according to Price's model of Section 14.1. Older vertices in the network—those added earlier in the growth process—have more time to acquire links from other vertices and hence we might expect that they would on average have higher in-degree. This indeed turns out to be the case, as we can show by calculating the degree distribution as a function of the time at which vertices are created.

Let $p_q(t, n)$ be the average fraction of vertices in our directed network that were created at time t and have in-degree q when the network has n vertices total. The time of creation is measured in terms of the number of vertices, the first vertex having $t = 1$ and the last having $t = n$. Alternatively, you can just think of t as counting the vertices from 1 to n, recording the order in which they were added. Strictly t need not reflect actual time, because the vertices need not have been added at a constant rate, but if we know the real times at which vertices were added we can easily convert between our timescale and real time.

We can write down a master equation for the evolution of $p_q(t,n)$ as follows. Upon the addition of a new vertex to the network, the expected number of new edges acquired by previously existing vertices with in-degree q is independent of time of those vertices creation and, following Eq. (14.2), is given by

$$np_q(t,n) \times c \times \frac{q+a}{n(c+a)} = \frac{c(q+a)}{c+a} p_q(t,n),$$

(14.41)

with the parameters c and a defined as in Section 14.1. Then the master equation takes the form

$$(n+1)p_q(t,n+1) = np_q(t,n) + \frac{c}{c+a}\left[(q-1+a)p_{q-1}(t,n) - (q+a)p_q(t,n)\right].$$

(14.42)

The only exception to this equation is, as before, for the case of $q = 0$, where we get

$$(n+1)p_0(t,n+1) = np_0(t,n) + \delta_{tn} - \frac{ca}{c+a} p_0(t,n).$$

(14.43)

Notice the Kronecker delta, which adds a single vertex of in-degree zero if $t = n$, but none otherwise.

These equations, though correct, don't make much sense in the limit of large n, since the fraction of vertices created at time t goes to zero in this limit because only one vertex is created at any particular t. So instead we change variables to a rescaled time

$$\tau = \frac{t}{n},$$

(14.44)

which takes values between zero (oldest vertices) and one (youngest vertices). At the same time we also change from $p_q(t, n)$ to a density function $\pi_q(\tau, n)$ such that $\pi_q(\tau, n)\, d\tau$ is the fraction of vertices that have in-degree q and fall in the interval from τ to $\tau + d\tau$. The number of vertices in the interval $d\tau$ is $n\, d\tau$, which implies that $\pi_q\, d\tau = p_q \times n\, d\tau$ and hence

$$\pi_q(\tau, n) = n p_q(t, n).$$

(14.45)

Being a density function, π_q does not vanish as $n \to \infty$.

The downside of this variable change is that τ is no longer constant for a given vertex. A vertex created at time t has rescaled time t/n when there are n vertices in the network but $t/(n + 1)$ when there are $n + 1$. Thus, in terms of τ and π_q, Eq. (14.42) becomes

$$\pi_q\left(\frac{n}{n+1}\tau, n+1\right) = \pi_q(\tau, n) + \frac{c}{c+a}\left[(q-1+a)\frac{\pi_{q-1}(t,n)}{n} - (q+a)\frac{\pi_q(t,n)}{n}\right].$$

(14.46)

Now we consider the limit where $n \to \infty$. If we define the shorthand notation $\pi_q(t) = \pi_q(t, \infty)$ and the small quantity $\epsilon = 1/n$, Eq. (14.42) becomes

$$\frac{\pi_q(\tau) - \pi_q(\tau - \epsilon\tau)}{\epsilon} + \frac{c}{c+a}\left[(q-1+a)\pi_{q-1}(\tau) - (q+a)\pi_q(\tau)\right] = 0,$$

(14.47)

where we have dropped terms of order ϵ^2.

As $n \to \infty$, we have $\epsilon \to 0$ and the first two terms become a derivative thus:

$$\lim_{\epsilon \to 0} \frac{\pi_q(\tau) - \pi_q(\tau - \epsilon\tau)}{\epsilon} = \tau \frac{d\pi_q}{d\tau},$$

(14.48)

and so our master equation becomes a differential equation in this case:

$$\tau \frac{d\pi_q}{d\tau} + \frac{c}{c+a}\left[(q-1+a)\pi_{q-1}(\tau) - (q+a)\pi_q(\tau)\right] = 0.$$

(14.49)

The corresponding equation for $q = 0$ is

$$\tau \frac{d\pi_0}{d\tau} - \frac{ca}{c+a}\pi_0(\tau) = 0,$$

(14.50)

so long as $\tau < 1$. For the special case $\tau = 1$ the δ_{tn} in Eq. (14.43) presents a problem, but on the other hand for $\tau = 1$ we know what the answer is anyway: there is always exactly one vertex created at time $t = n$, which has in-degree zero, so $\pi_0(1) = 1$ in the language of our rescaled variables.[213] In effect, this just provides a boundary condition on $\pi_0(\tau)$. The corresponding boundary condition for $q \geq 1$ is $\pi_q(1) = 0$, since there are no vertices with $t = n$ and $q \geq 1$.

We can solve Eqs. (14.49) and (14.50) by starting with a solution for $q = 0$ and working up through increasing values of q. This is similar to our solution for the degree distribution, Eq. (14.21), except that the equations we are solving are now differential equations.

The solution for the $q = 0$ case is straightforward—Eq. (14.50) is homogeneous in π_0 and can be solved by standard methods. You can easily verify that the solution is $\pi_0(\tau) = A\tau^{ca/(c+a)}$ where A is an integration constant. The constant is fixed by the boundary condition $\pi_0(1) = 1$, which implies that $A = 1$ and hence

$$\pi_0(\tau) = \tau^{ca/(c+a)}.$$

(14.51)

As a check, we can integrate over τ to get the total fraction of vertices with in-degree zero:

$$\int_0^1 \tau^{ca/(c+a)}\,d\tau = \frac{1+a/c}{a+1+a/c},$$

(14.52)

which agrees nicely with our previous result for the same quantity, Eq. (14.9).

Now we can use this solution to find $\pi_1(\tau)$. Equation (14.49) tells us that

$$\tau\frac{d\pi_1}{d\tau} - \frac{c}{c+a}(a+1)\pi_1(\tau) = -\frac{c}{c+a}\,a\pi_0(\tau) = -\frac{ca}{c+a}\tau^{ca/(c+a)}.$$

(14.53)

This is again just an ordinary first-order differential equation, although an inhomogeneous one this time (i.e., it has a driving term on the right-hand side).

We tackle it in standard fashion. First we find the general solution for the homogeneous equation in which the right-hand side is set to zero, which is $B\tau^{c(a+1)/(c+a)}$ where B is an integration constant. Then we find any (nongeneral) solution to the full equation with the driving term included—the obvious one is $a\tau^{ca/(c+a)}$—and sum the two. The constant is fixed by the boundary condition $\pi_1(1) = 0$, which implies that $B = -a$, and we get

$$\pi_1(\tau) = a\tau^{ca/(c+a)}\bigl(1 - \tau^{c/(c+a)}\bigr).$$

(14.54)

Now, by a similar method, we can use *this* solution to solve for $\pi_2(\tau)$, and so forth to higher and higher values of q. The algebra is tedious, but with persistence you can show that the next two results are

$$\pi_2(\tau) = \tfrac{1}{2}a(a+1)\tau^{ca/(c+a)}\bigl(1 - \tau^{c/(c+a)}\bigr)^2,$$

(14.55)

$$\pi_3(\tau) = \tfrac{1}{6}a(a+1)(a+2)\tau^{ca/(c+a)}\bigl(1-\tau^{c/(c+a)}\bigr)^3.$$

(14.56)

These results suggest the general solution (first given by Dorogovtsev *et al.* [99])

$$\begin{aligned}\pi_q(\tau) &= \frac{1}{q!}\bigl[a(a+1)\ldots(a+q-1)\bigr]\tau^{ca/(c+a)}\bigl(1-\tau^{c/(c+a)}\bigr)^q \\ &= \frac{\Gamma(q+a)}{\Gamma(q+1)\Gamma(a)}\tau^{ca/(c+a)}\bigl(1-\tau^{c/(c+a)}\bigr)^q,\end{aligned}$$

(14.57)

where we have made use again of the convenient property of the gamma function derived in Eq. (14.17) as well as the result that $\Gamma(n+1) = n!$ when n is a positive integer.[214] With a little work you can verify that this is indeed a complete solution of Eq. (14.49) for all q. As a check we can also integrate over τ to find the total fraction of vertices with in-degree q and confirm that the result agrees with Eq. (14.21). We leave this calculation as an exercise for the reader.[215]

Let us take a moment to examine the structure of our solution for $\pi_q(\tau)$ and see what it tells us about the network. The general shape of the solution is shown in Fig. 14.3. Panel (a) shows the distribution of creation times τ for vertices of given in-degree q for various values of q and for each value there is a clear peak in the distribution, indicating that vertices of a given degree are concentrated around a particular era in the growth of the graph. As degree increases, that era gets earlier, so that the times of creation of vertices that ultimately achieve high in-degree are strongly concentrated around the beginning of the growth process.

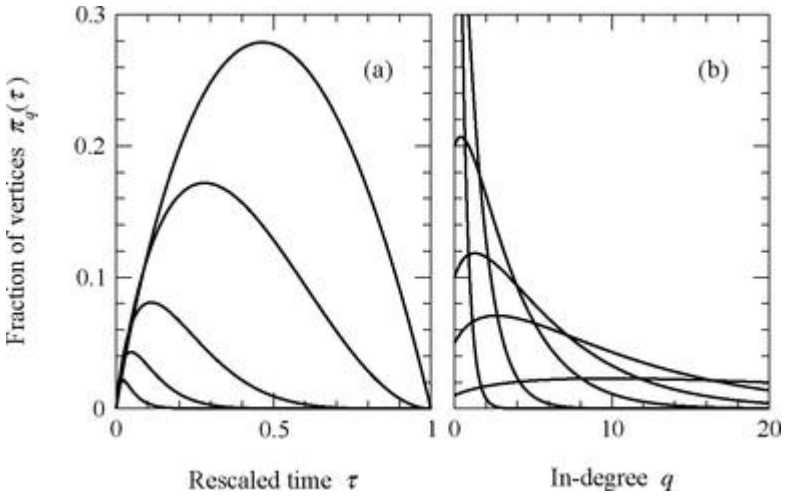


Figure 14.3: Distribution of vertices in Price's model as a function of in-degree and time of creation. The two panels show the distribution $\pi_q(\tau)$, Eq. (14.57), for $c = 3$ and $a = 1.5$ as (a) a function of τ for (top to bottom) $q = 1$, 2, 5, 10, and 20, and (b) a function of q for $\tau = 0.01$ (flattest curve), 0.05, 0.1, 0.5, and 0.9 (steepest curve).

Panel (b) of Fig. 14.3 shows the distribution of in-degrees for vertices created at a selection of different times τ. This distribution also has a peak, then falls off sharply as q becomes large.[216] Indeed the distribution falls off roughly exponentially as q becomes large, as we can see from Eq. (14.57) by writing

$$\frac{\Gamma(q+a)}{\Gamma(q+1)\Gamma(a)} = \frac{\Gamma(q+a)}{q\Gamma(q)\Gamma(a)} = \frac{1}{q\mathrm{B}(q,a)},$$

(14.58)

where we have used Euler's beta function again, Eq. (14.19). As shown in Section 14.1, the beta function has a power-law tail $\mathrm{B}(x,y) \sim x^{-y}$ for large x (Eq. (14.25)) so π_q goes with q as

$$\pi_q(\tau) \sim q^{a-1}\bigl(1-\tau^{c/(c+a)}\bigr)^q.$$

(14.59)

In other words it decays exponentially except for a leading algebraic factor. Thus the degree distribution for vertices with specific values of τ does not follow a power law. The power-law behavior seen in the full degree distribution of the model, Eq. (14.21), only appears when we integrate over all times τ. However, the decay of the exponential in Eq. (14.59) is slower for smaller τ, so older vertices are more likely to have high in-degree than younger ones, as we saw in Fig. 14.3.

To investigate this point further, we can calculate the mean in-degree $\gamma(\tau)$ for a vertex created at time τ thus:

$$\gamma(\tau) = \sum_{q=0}^{\infty} q\pi_q(\tau) = a\bigl(\tau^{-c/(c+a)} - 1\bigr).$$

(14.60)

Figure 14.4 shows the shape of $\gamma(\tau)$ for a variety of choices of the parameters and, as we expect, the mean value of the in-degree increases with decreasing τ and eventually diverges as τ approaches zero. Notice, though, that no vertex ever actually has $\tau = 0$. The first vertex added to the network has $t = 1$, so the smallest value of τ is $1/n$. Nonetheless, we see that vertices added to the network early have an enormous advantage in terms of in-degree over those added even a little

later. For a citation network, for instance, this suggests that the early papers in a field will receive substantially more citations than later ones, purely because they were published first.

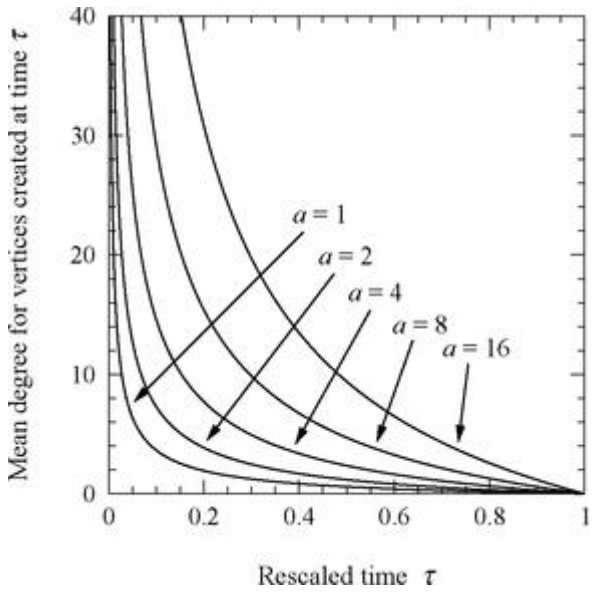


Figure 14.4: Average in-degree of vertices as a function of their time of creation. The average in-degree of vertices in Price's network model as a function of the rescaled time $\tau = t/n$ at which they were added to the network, in the limit of large n for various values of the parameter a. The out-degree parameter c was in each case $c = 2a$, so that the exponent of the power-law degree distribution $\alpha = 2 + a/c$ (Eq. (14.27)) is 2.5 for all curves, which is a typical value for real-world networks.

Indeed, this is a pattern seen in many different areas, not just in networks. In any situation where success begets more success, first movers are expected to have a large advantage over others. Any small lead gained early in the process is quickly amplified by the preferential attachment process into a bigger lead and soon the lucky first movers find themselves racing ahead of the pack. Those who enter the game later may experience chance fluctuations that give them a small boost, but since there are probably many others already ahead of them, that boost is not amplified significantly because most of the wealth is already going to the leaders under the preferential attachment rule.

A nice demonstration of this process, although not in the field of networks, has been given by Salganik *et al.* [288], who examined the behavior of a group of people downloading popular music online. Salganik *et al.* created a website on which participants could download and listen to songs by little-known artists for free. Participants were told how many times each song had previously been downloaded and Salganik and coworkers found that there was a clear preferential attachment effect: songs with many previous downloads were downloaded far more than those with few. As a result there was a strong first-mover advantage, with songs that took an early lead benefiting from the preferential attachment and turning that lead into a much larger one, resulting in a roughly power-law distribution in the numbers of downloads.

To test the theory that they were seeing a preferential attachment process rather than actual differences in song quality leading to different download rates, Salganik *et al.* then changed the download numbers reported for each song, deliberately misrepresenting the number of times each had been downloaded. They discovered when they did this that the songs with the highest *reported* numbers of downloads were still downloaded most often, even though the reported numbers no longer corresponded to true popularity.[217] These results strongly suggest that success is, at least in this context and at least in part, a result of previous success and that a good way to be successful is

to get in at the beginning and get an early lead. Of course, that may be easier said than done. Many people would like to get in at the beginning of a new field of scientific research or a new business opportunity, but it's not always clear how one should do it.

Returning to our network growth model, it is also interesting to ask how the expected in-degree of a vertex varies with its age after it enters the network. This differs from the expected degree for a particular τ calculated above because a given vertex does not have a fixed value of τ. The value of $\tau = t/n$ for a vertex decreases as time passes because n is increasing. For this reason the behavior of individual vertices is more easily understood in terms of our original non-rescaled time t, which does remain constant.

So let t again be the time at which a vertex is added to the network and let s be the subsequent elapsed time, i.e., the age of the vertex. Necessarily we have $s + t = n$ and hence

$$\tau = \frac{t}{n} = \frac{t}{t+s}.$$

(14.61)

Substituting this expression into Eq. (14.60), we then find the expected indegree $\gamma_t(s)$ of the vertex added at time t, as a function of its age s, to be

$$\gamma_t(s) = a\left[\left(1+\frac{s}{t}\right)^{c/(c+a)} - 1\right].$$

(14.62)

When a vertex is first added to the network and $s \ll t$, we can expand in the small quantity s/t to give

$$\gamma_t(s) \simeq \frac{ca}{c+a}\left(\frac{s}{t}\right).$$

(14.63)

In other words, the in-degree of a vertex initially grows linearly with the age of the vertex, on average, but with a constant of proportionality that is smaller the later the vertex entered the network—again we see that there is a substantial advantage for vertices that enter early.

As the vertex ages, there is a crossover to another regime around the point $s = t$, i.e., at the point where the vertex switches from being in the younger half of the population to being in the older. For $s \gg t$,

$$\gamma_t(s) \simeq a\left(\frac{s}{t}\right)^{c/(c+a)},$$

(14.64)

which has a similar form to Eq. (14.63) but with a different exponent, $c/(c + a)$, which is always less than 1, so the growth is slower than linear but still favors vertices that appear early. Figure 14.5 shows the behavior of $\gamma_t(s)$ with time for vertices created at a selection of different times t.

All of these results can be applied to the Barabási-Albert model as well by setting $a = c$ with c an integer and writing the formulas in terms of total degree $k = q + c$ rather than in-degree. For instance, the joint degree/time distribution, Eq. (14.57), becomes

$$\pi_k(\tau) = \binom{k-1}{k-c}\sqrt{\tau}(1-\sqrt{\tau})^{k-c}$$

(14.65)

for $k \geq c$ and $\pi_k(\tau) = 0$ for $k < c$. This result was first given by Krapivsky and Redner [187] for the case $c = 1$.

14.3.2 SIZES OF IN-COMPONENTS

The in-components of vertices in our growing networks have some interesting properties. Recall that the in-component of vertex i is the set of vertices from which i can be reached by following a directed path through the network (see Section 6.11). In a citation network, for instance, the in-component of paper A is the set of all papers from which A can be reached by following some trail of successive citations. The reader reading any paper in the in-component can look up other papers in its bibliography, find those papers and look up further ones in *their* bibliographies and so forth, and ultimately reach paper A. One can think of the in-component as representing the set of all papers that "indirectly cite" paper A and the size of the in-component can be considered a measure of the total impact of paper A.

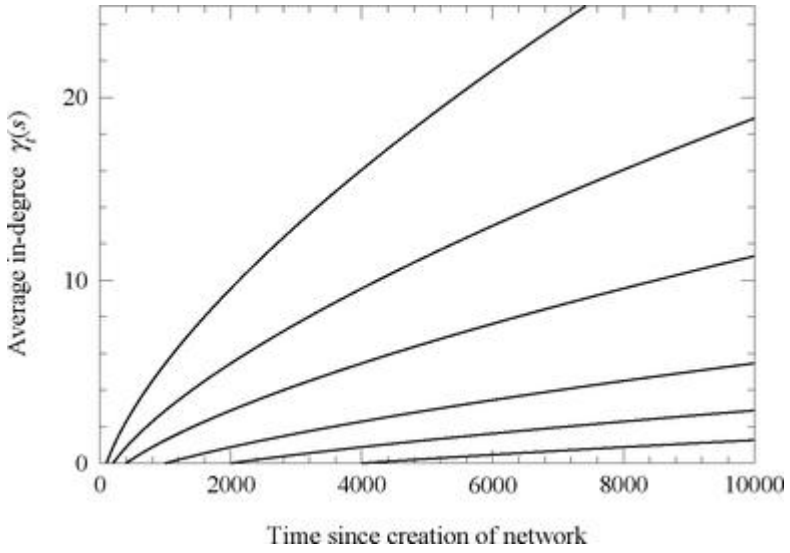


Figure 14.5: Average in-degree of vertices created at different times. The curves show the average in-degrees in Price's model of vertices created at times (top to bottom) t = 100, 200, 400, 1000, 2000, and 4000 as a function of time since the creation of the network. The model parameters were $c = 3$ and $a = 1.5$.

We can study the distribution of in-component sizes by a method similar to the one we used for the degree distribution. Consider Fig. 14.6, which shows a sketch of the in-component of a single vertex A. We have drawn the in-component as a tree, which is accurate so long as the size s of the component is small, $s \ll n$. Just as in the random graph models of Chapters 12 and 13, the probability of a small component having an extra edge that destroys its tree structure vanishes in the limit of large n (see Section 12.6.1). We must be careful however. As we will shortly see, it is possible for the sizes of in-components to become comparable with n in preferential attachment models, in which case the arguments below break down. For the moment, however, let us proceed under the assumption that our component is a tree.

Our in-component will grow in size as vertices and edges are added to the network. Specifically, it will grow larger every time a newly added edge links to any of its members. The probability of a new edge linking to vertex i is given by Eq. (14.1) and summing this probability over all vertices in the in-component C gives a total probability of

$$\sum_{i\in\mathcal{C}} \frac{q_i + a}{n(c+a)} = \frac{1}{n(c+a)}\left[as + \sum_{i\in\mathcal{C}} q_i\right],$$

(14.66)

where s is the number of vertices in the in-component.

Considering Fig. 14.6, we see that every incoming edge in the in-component is necessarily also an outgoing edge from another vertex in the in-component, and moreover that there is exactly one such outgoing edge from each vertex in the in-component, except for A itself. (There are also other outgoing edges from vertices in the in-component, as shown in gray in the figure, but these connect to vertices outside the component itself and play no part in our calculation.) Thus the total number $\sum_{i\in C} q_i$ of incoming edges is equal to the number of vertices in the in-component minus one, which means our probability of connection, Eq. (14.66), can also be written as

$$\frac{1}{n(c+a)}(as + s - 1) = \frac{(1+a)s - 1}{n(c+a)}.$$

(14.67)

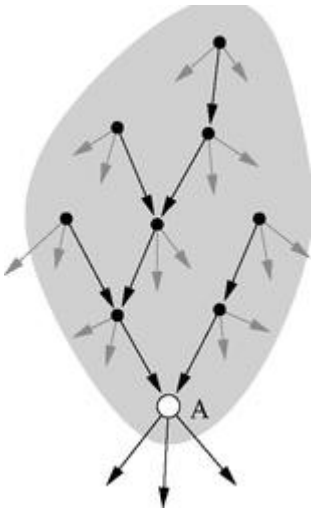


Figure 14.6: The in-component of a vertex A. The total number of incoming links attached to vertices in an in-component is equal to the number of vertices in the component minus 1. Note that there are, in general, many edges outgoing from vertices in the in-component (shown in gray) which connect to vertices not in the in-component and which can thus be ignored for the purposes of our calculation.

Let us define $p_s(n)$ to be the probability that a randomly chosen vertex has an in-component of size s when the network has n vertices (still assuming $s \ll n$). Then $np_s(n)$ is the number of in-

components of size s and, given that each new vertex arrives with c outgoing links, the total number of in-components of size s receiving a new link upon the addition of a new vertex to the network is

$$np_s(n) \times c \times \frac{(1+a)s-1}{n(c+a)} = \frac{c}{c+a}[(1+a)s-1]p_s(n).$$

(14.68)

Now, by an argument similar to the one used to derive Eq. (14.5), we can show that $p_s(n)$ satisfies the master equation

$$(n+1)p_s(n+1) = np_s(n) + \frac{c+ca}{c+a}\left[\left(s-1-\frac{1}{1+a}\right)p_{s-1}(n) - \left(s-\frac{1}{1+a}\right)p_s(n)\right].$$

(14.69)

The only exception is for in-components of size 1, the smallest size possible, for which

$$(n+1)p_1(n+1) = np_1(n) + 1 - \frac{ca}{c+a}p_1(n).$$

(14.70)

The +1 here represents the fact that there is one new in-component of size 1 created for each vertex added.

We now take the limit $n \to \infty$ and write $p_s = p_s(\infty)$ to get

$$p_s = \frac{c+ca}{c+a}\left[\left(s-1-\frac{1}{1+a}\right)p_{s-1} - \left(s-\frac{1}{1+a}\right)p_s\right]$$

(14.71)

for $s \geq 2$ and

$$p_1 = 1 - \frac{ca}{c+a}p_1.$$

(14.72)

The solution of these equations follows exactly the same lines as the solution for degree distribution. The final result is

$$p_s = \frac{\mathrm{B}(s - 1/(1+a), \beta)}{\mathrm{B}(1 - 1/(1+a), \beta - 1)},$$

(14.73)

where B(x,y) is the Euler beta function, Eq. (14.19), and

$$\beta = 1 + \frac{1 + a/c}{1 + a}.$$

(14.74)

As we have seen previously, the beta function has a power-law tail B(x,y) ~ x^{-y} (see Eq. (14.25)), so the in-component size distribution also has a power-law tail:

$$p_s \sim s^{-\beta},$$

(14.75)

although with an exponent β that is in general different from that of the degree distribution (see Eq. (14.27)). Indeed, note that for the normal situation where $c \geq 1$ we have $1 < \beta \leq 2$, which is puzzling: power laws with $\beta \leq 2$ have no finite mean, but for any finite value of n our network must certainly have a finite average component size. The solution to this conundrum is relatively simple, however. As we pointed out above, our calculations are only valid for component sizes $s \ll n$. For larger sizes the method will break down and the power-law behavior will be lost. In physical terms, in-components clearly cannot be larger than the size of the whole network, and so we must expect finite-size effects that cut off the size distribution as s approaches n. In mathematical terms, the components stop being trees as their size becomes comparable with n and hence Eq. (14.67) ceases to be correct.

See Section 8.4.2 for a discussion of the mean and other moments of power-law distributions.

It's also possible to derive solutions for the component size distribution as a function of the time of creation of a vertex or the age of a vertex, just as we did for the degree distribution in the first

part of this section. Indeed there are many more properties of these models that can be calculated using the master equation approach, which is an immensely useful technique for problems such as these. Life, however, is short, and there are many other interesting matters to look into, so we will move on to other things.

14.4 EXTENSIONS OF PREFERENTIAL ATTACHMENT MODELS

Many extensions and generalizations of preferential attachment models have been suggested, typically addressing questions about what happens when we vary the details of the model definition or attempt to make the model more faithful to the way real networks grow. For instance:

1. By contrast with citations, links in the Web are not permanent. They can and frequently do disappear as well as appear, and links can be added between vertices not just at the moment a vertex is created but at any later time too.
2. Entire web pages also disappear as well as appear.
3. There is no obvious reason why the preferential attachment process has to be linear in the degree. What happens if it is non-linear?
4. Not all vertices are created equal. Some papers or websites might be intrinsically more interesting or important by virtue of their content and hence attract more links. Can this process be incorporated into the model?

In this section we describe modifications of the preferential attachment process that address each of these questions. In the interests of simplicity, we describe the developments in the context of the Barabási-Albert model, rather than the more general Price model. Generalizations of Price's model are certainly possible but the algebra is in many cases unwieldy, and the main conclusions are easier to understand in the context of the simpler model.

14.4.1 ADDITION OF EXTRA EDGES

Price proposed his model of a growing network with citation networks in mind. Since the bibliography of a paper cannot be changed after the paper is published, the edges in a citation network are effectively frozen in place from the moment they are first created, and Price's model mimics this behavior with edges being added only at the moment a vertex is created and never moved or removed thereafter.

This is not true of all networks, however. The World Wide Web, for example, is constantly changing. Links between web pages can, and often are, added or removed after the pages are created. This state of flux is not captured by Price′s model or by the Barabási-Albert model of Section 14.2. Yet the Web still has a power-law degree distribution. This leads us to wonder whether it is possible to create a generalized model that includes the addition and removal of edges after vertices are created but still generates power-law distributions. It turns out that we can indeed, as we now describe.

We first consider the relatively simple case in which edges are added to our network but never taken away, which has been studied by a number of authors [11, 96, 190]. The case of edge removal is more complex and is considered in the following section. The model we consider is a generalization of the Barabási-Albert model in which vertices are added to the network one by one as before and each starts out with c undirected edges which attach to other vertices with probability proportional to degree k. But we now include a second process in the model as well: at each step some number w of extra edges are added to the network with *both* ends attaching to vertices chosen in proportional to degree. Thus when the network has n vertices it will have a total of $n(c + w)$ edges. (In fact, it is only necessary that an *average* number w of extra edges be added at each step. The actual number can fluctuate around this figure, provided the fluctuations satisfy some modest constraints on their size, and the net result, in the limit of large network size, will be the same. This allows us to give w a non-integer value if we wish.)

This model turns out to be quite easy to solve given the results of previous sections. The only difference between it and the standard Barabási-Albert model is that, instead of c new ends of edges attaching to old vertices for every new vertex added, we now have $c + 2\omega$new ends of edges—two extra for each of the w extra edges. The probability of attachment of any one of those edges to a particular vertex i is $k_i / \sum_i k_i$ as before. The sum in the denominator is equal to twice the number of edges in the network (see Eq. (6.20)), or $\sum_i k_i = 2n(c + w)$.

Then, if $p_k(n)$ denotes the fraction of vertices with degree k when the network has n vertices in total, the number of vertices of degree k receiving a new edge, per vertex added, is

$$np_k(n) \times (c + 2w) \times \frac{k}{2n(c + w)} = \frac{c + 2w}{2(c + w)} kp_k(n).$$

(14.76)

We can use this result to write a master equation for $p_k(n)$ thus:

$$(n + 1)p_k(n + 1) = np_k(n) + \frac{c + 2w}{2(c + w)} [(k - 1)p_{k-1}(n) - kp_k(n)],$$

(14.77)

for $k > c$ and

$$(n+1)p_c(n+1) = np_c(n) + 1 - \frac{c+2w}{2(c+w)}cp_c(n),$$

(14.78)

for $k = c$. (There are, as before, no vertices of degree less than c.) Taking the limit of large n and writing $p_k = p_k(\infty)$, these equations simplify to

$$p_k = \frac{c+2w}{2(c+w)}[(k-1)p_{k-1} - kp_k] \qquad \text{for } k > c,$$

(14.79)

$$p_c = 1 - \frac{c+2w}{2(c+w)}cp_c \qquad \text{for } k = c.$$

(14.80)

Rearranging these equations along the lines of Eqs. (14.9) to (14.21), we then find that

$$p_k = \frac{\mathrm{B}(k,\alpha)}{\mathrm{B}(c,\alpha-1)},$$

(14.81)

where $\mathrm{B}(x,y)$ is the Euler beta function again, Eq. (14.19), and

$$\alpha = 2 + \frac{c}{c+2w}.$$

(14.82)

Since $B(x,y)$ goes as x^{-y} for large x (Eq. (14.25)), our degree distribution has a power-law tail with exponent α. For the special case of $w = 0$, in which no additional edges are added to the network, we recover the standard result $\alpha = 3$ for the Barabási-Albert model; for $\omega > 0$ we get exponents in the range $2 < \alpha < 3$, which agrees nicely with the values typically observed for degree distributions on the Web (see Table 8.1). Bear in mind though that the Web is a directed network while the model described here is undirected. If we want to build a model of a directed network we would need to start with something like the Price model of Section 14.1. Generalizations of Price's model that include addition of extra edges as above are certainly possible—see for example Krapivsky *et al.* [190].

14.4.2 REMOVAL OF EDGES

Now consider the case of a network in which edges can be removed. To keep things simple let us first consider the case where edges can be removed at any time but are only added at the initial creation of a vertex, as in the standard Barabási-Albert model. (In a moment we will consider the general case of addition and removal at any time.)

There are many ways in which edges could be removed from a network, but let us consider the most basic case in which they are simply deleted uniformly at random. What then is the probability that a particular vertex i loses an edge when a single edge is removed from the network? When an edge is deleted both of its two ends vanish. Given that the deletion is uniformly random, the probability that one of those two ends is attached to vertex i is simply proportional to the total number of ends attached to i, which is equal to the degree k_i. Properly normalized, the probability that vertex i loses an edge is thus $2k_i / \sum_i k_i$, the factor of two coming from the two ends of the edge. In other words, the random deletion of edges is like a type of preferential attachment in reverse: the higher the degree of the vertex, the more likely it is to lose an edge.

So consider the undirected network model in which vertices with degree c are added to the network following the normal preferential attachment scheme and an average of v edges are deleted at random for each vertex added. (As with the model of Section 14.4.1 the actual number of edges deleted can fluctuate about the mean and v can take a non-integer value if we wish.) In order that the number of edges in the network grow, rather than shrinking to zero and vanishing, we require that the net number of edges added per vertex $c - v$ be positive, i.e., $v < c$. Then when the network has n vertices the number of edges will be $n(c - v)$.

In writing down a master equation for this model there are several processes we need to consider. As before, the number of vertices with degree k increases whenever a vertex of degree $k - 1$ gains a new edge and decreases when a vertex of degree k gains a new edge. By an argument analogous to the one leading to Eq. (14.76), the number of vertices of degree k gaining an edge per vertex added to the network is

$$np_k(n) \times c \times \frac{k}{2n(c-v)} = \frac{c}{2(c-v)} k p_k(n).$$

(14.83)

But we also now have a new process in which a vertex can lose an edge, which means that the number of vertices of degree k also increases when a vertex of degree $k + 1$ loses an edge and decreases when a vertex of degree k loses an edge. The number of vertices of degree k losing an edge per vertex added is given by

$$np_k(n) \times 2v \times \frac{k}{2n(c-v)} = \frac{v}{c-v} k p_k(n),$$

(14.84)

with the factor of $2v$ reflecting the fact that each of the v edges removed has two ends.

Another important thing to notice is that, by contrast with the original Barabási-Albert model, vertices can now have any degree $k \geq 0$—vertices can lose any or all of their edges, right down to the last one, so there is no restriction $k \geq c$ on the degree as before.

Our master equation now takes the form

$$(n+1)p_k(n+1) = np_k(n) + \frac{c}{2(c-v)}(k-1)p_{k-1}(n) + \frac{v}{c-v}(k+1)p_{k+1}(n) - \frac{c+2v}{2(c-v)}kp_k(n)$$

(14.85)

for $k \neq c$ and

$$(n+1)p_c(n+1) = np_c(n) + 1 + \frac{c}{2(c-v)}(c-1)p_{c-1}(n) + \frac{v}{c-v}(c+1)p_{c+1}(n) - \frac{c+2v}{2(c-v)}cp_c(n)$$

(14.86)

for $k = c$. These two equations can conveniently be combined by writing

$$(n+1)p_k(n+1) = np_k(n) + \delta_{kc} + \frac{c}{2(c-v)}(k-1)p_{k-1}(n) + \frac{2v}{2(c-v)}(k+1)p_{k+1}(n) - \frac{c+2v}{2(c-v)}kp_k(n),$$

(14.87)

where δ_{kc} is the Kronecker delta, which is 1 if $k = c$ and 0 otherwise.

The only exception to this master equation is for the case $k = 0$, where the term proportional to $k - 1$ vanishes because there are no vertices of degree -1. A simple way of enforcing this exception is ≥ to define $p_{-1}(n) = 0$ for all n, in which case Eq. (14.87) then applies for all $k \geq 0$. We will adopt this convention henceforth.

The model as we have described it so far incorporates the processes of vertex addition and edge removal, but, given Eq. (14.87), it is only a small extra step to incorporate the edge addition process of Section 14.4.1 as well. If as before we add ω extra edges per vertex added, then $c + \omega - v$ edges are added net per vertex, and our master equation becomes

$$(n+1)p_k(n+1) = np_k(n) + \delta_{kc} + \frac{c+2w}{2(c+w-v)}(k-1)p_{k-1}(n) + \frac{v}{c+w-v}(k+1)p_{k+1}(n) - \frac{c+2w+2v}{2(c+w-v)}kp_k(n).$$

(14.88)

The equation for edge removal only, Eq. (14.87), can then be considered a special case of this equation with $w = 0$. As before, we require that the net number of edges added per vertex be positive, or $v < c + w$.

Now taking the limit as $n \to \infty$ and writing $p_k = p_k(\infty)$ we find that

$$p_k = \delta_{kc} + \frac{c+2w}{2(c+w-v)}(k-1)p_{k-1} + \frac{v}{c+w-v}(k+1)p_{k+1} - \frac{c+2w+2v}{2(c+w-v)}kp_k.$$

(14.89)

This equation differs in a crucial way from the master equations we have encountered previously, such as Eq. (14.7), because the right-hand side contains terms for vertices of three different degrees ($k - 1$, k, and $k + 1$) rather than just two. This makes the equation substantially more difficult to solve. We can no longer simply rearrange to derive an expression for p_k in terms of p_{k-1} and then apply that expression repeatedly to itself. A solution is still possible, but it's not simple. Here we give just an outline of the method. The gory details, for those interested in them, are spelled out by Moore *et al.* [226].[218]

The basic strategy for solving Eq. (14.89) is to use a generating function of the kind we introduced in Section 13.1. We define

$$g(z) = \sum_{k=0}^{\infty} p_k z^k.$$

(14.90)

Substituting for p_k from Eq. (14.89) we get

$$g(z) = \sum_{k=0}^{\infty} \delta_{kc} z^k + \frac{c+2w}{2(c+w-v)} \sum_{k=0}^{\infty} (k-1)p_{k-1} z^k + \frac{2v}{2(c+w-v)} \sum_{k=0}^{\infty} (k+1)p_{k+1} z^k - \frac{c+2w+2v}{2(c+w-v)} \sum_{k=0}^{\infty} kp_k z^k.$$

(14.91)

The first term on the right is simple—it is equal to z^c . The others require a little more care. Consider the second term, for example. Note that the first term in the sum, the term for $k = 0$, is necessarily zero because, as we have said, $p_{-1} = 0$. Hence we can write

$$\begin{aligned}\sum_{k=0}^{\infty}(k-1)p_{k-1}z^k &= \sum_{k=1}^{\infty}(k-1)p_{k-1}z^k = \sum_{k=0}^{\infty}kp_kz^{k+1}\\ &= z^2\sum_{k=0}^{\infty}kp_kz^{k-1} = z^2\frac{\mathrm{d}}{\mathrm{d}z}\sum_{k=0}^{\infty}p_kz^k\\ &= z^2\frac{\mathrm{d}g}{\mathrm{d}z},\end{aligned}$$

(14.92)

where in the first line we have made the substitution $k - 1 \to k$ and in the second line we have made use of the fact that the $k = 0$ term is again zero (because of the factor of k).

For the third and fourth terms in (14.91) we can similarly write

$$\sum_{k=0}^{\infty}(k+1)p_{k+1}z^k = \sum_{k=1}^{\infty}kp_kz^{k-1} = \sum_{k=0}^{\infty}kp_kz^{k-1} = \frac{\mathrm{d}g}{\mathrm{d}z},$$

(14.93)

and

$$\sum_{k=0}^{\infty}kp_kz^k = z\sum_{k=0}^{\infty}kp_kz^{k-1} = z\frac{\mathrm{d}g}{\mathrm{d}z}.$$

(14.94)

Combining Eqs. (14.91) to (14.94) and rearranging, we then get

$$\frac{(c+2w)z-2v}{2(c+w-v)}(1-z)\frac{\mathrm{d}g}{\mathrm{d}z} + g(z) = z^c.$$

(14.95)

This is a first-order linear differential equation and is solvable by standard—if tedious—methods. To cut a long story short, one can find an integrating factor for the left-hand side and hence express the solution in terms of an integral that, provided $v < \frac{1}{2}c + w$, can be reduced by repeated integration by parts to give

$$p_k = Ak^{-\alpha} \int_0^k \frac{(1 - x/k)^k}{(1 - \gamma x/k)^k} x^{\alpha - 2} \, dx,$$

(14.96)

for $k \geq c$, where A is a k-independent constant and

$$\alpha = 2 + \frac{v - w}{c + 2w - 2v},$$

(14.97)

$$\gamma = \frac{2v}{c + 2w}.$$

(14.98)

The remaining integral can be written in terms of hypergeometric functions, but we can find the asymptotic behavior of the degree distribution for large k more directly by noticing that as k becomes large

$$\left[1 - \frac{x}{k}\right]^k \to e^x, \qquad \left[1 - \frac{\gamma x}{k}\right]^k \to e^{\gamma x},$$

(14.99)

so that

$$p_k \sim k^{-\alpha} \int_0^\infty \mathrm{e}^{-(1-\gamma)x} x^{\alpha-2}\,\mathrm{d}x = \frac{\Gamma(\alpha-1)}{(1-\gamma)^{\alpha-1}}\, k^{-\alpha}.$$

(14.100)

Thus we once again find that our degree distribution has a power-law tail, with an exponent given this time by Eq. (14.97). Note that this exponent can take values both greater than and less than two. What's more for the case where $v = \frac{1}{2}c + w$ it actually becomes infinite. Moore *et al.* [226] show that at this point we lose the power-law behavior and the distribution becomes instead a stretched exponential. Up until this point, however, the distribution still follows a power law, albeit with a very large exponent as v grows larger. For values of $v > \frac{1}{2}c + w$, the solution becomes nonsensical, with a negative value of α, and one must return to the original differential equation (14.95) to find the solution for this case. We leave the developments, however, as an exercise for the especially avid reader.

Before we leave this topic, however, let us point out that the methods used to solve Eq. (14.89) can also be used to calculate what happens when we remove not edges but vertices from our network. Loss of vertices does occur in some networks, such as the World Wide Web, so it is potentially of interest to ask what effect it has on the degree distribution. In fact, the solution for this case is very similar to the solution for loss of edges, with a power-law distribution and an exponent that depends on the vertex loss rate, diverging as the rate of loss approaches the rate at which vertices are added. The details can be found in [226].

14.4.3 NON-LINEAR PREFERENTIAL ATTACHMENT

In the models we have considered so far, the probability that a new edge attaches to a vertex is linear in the degree of the vertex. Although this is a reasonable first guess about the way things might be, it's certainly also possible that attachment processes might not be linear. Indeed, there is some empirical evidence that this is the case. For instance, Jeong *et al.* [165] looked at the growth of several real-world networks, measuring the rate at which nodes acquired new edges. To avoid problems associated with the fact that the rate can depend not only on degree but also on the total size n of the network (see Eq. (14.1)), they restricted their observations to relatively short intervals of time. The measured rates, plotted as a function of vertex degree, showed that for some networks there was a roughly linear preferential attachment effect, but for others attachment appeared to be non-linear, going as some power γ of the degree with γ being significantly different from 1. (The values they observed were around $\gamma = 0.8$.)

What effect would non-linear preferential attachment have on the degree distribution of our network? Should we still expect to see power-law behavior in the non-linear case? The answers to these questions depend on the particular functional form of the attachment probability and there are an infinite variety of functional forms. We will look at some specific examples shortly, but for the moment let us keep the discussion completely general. Following an approach introduced by Krapivsky *et al.* [189], we define an *attachment kernel*, denoted a_k, which specifies the functional form of the attachment probability. For the model of Barabási and Albert, where attachment is simply proportional to degree, the attachment kernel would be $a_k = k$. For the non-linear attachment observed by Jeong *et al.* and discussed above, it would be $a_k = k^\gamma$. Note that the attachment kernel is not a probability, merely a functional form. The correctly normalized probability that a newly added edge attaches to a specific vertex i having degree k_i is $a_k / \sum_i a_{k_i}$.

So consider again a growing undirected network of the type discussed in previous sections and let $p_k(n)$ be the fraction of vertices with degree k when the network has n vertices. As before, an average of c new edges are added to the network with each new vertex, but preferential attachment is now non-linear, governed by the attachment kernel a_k, which means that, by analogy with Eq. (14.2), the expected number of vertices of degree k receiving a new connection when a single new vertex is added to the network is

$$np_k(n) \times c \times \frac{a_k}{\sum_i a_{k_i}} = \frac{c}{\mu(n)} a_k p_k(n),$$

(14.101)

where

$$\mu(n) = \frac{1}{n} \sum_{i=1}^{n} a_{k_i} = \sum_k a_k p_k(n).$$

(14.102)

Now the master equation for $p_k(n)$ is

$$(n+1)p_k(n+1) = np_k(n) + \frac{c}{\mu(n)}\left[a_{k-1}p_{k-1}(n) - a_k p_k(n)\right].$$

(14.103)

As before the term in $p_{k-1}(n)$ represents new vertices of degree k created when vertices of degree $k-1$ receive new edges and the last term in $p_k(n)$ represents vertices of degree k lost when they gain new edges to become vertices of degree $k+1$.

The only exception to this equation is for vertices of degree c, for which

$$(n+1)p_c(n+1) = np_c(n) + 1 - \frac{c}{\mu(n)}a_c p_c(n).$$

(14.104)

(And there are no vertices of degree less than c, since all vertices are created with degree c initially and edges are never removed.)

Taking the limit as $n \to \infty$ and writing $p_k = p_k(\infty)$ and $\mu = \mu(\infty)$, these equations become

$$p_k = \frac{c}{\mu}\left[a_{k-1}p_{k-1} - a_k p_k\right]$$

(14.105)

for $k > c$ and

$$p_c = 1 - \frac{ca_c}{\mu}p_c.$$

(14.106)

Note that μ depends via Eq. (14.102) on the degree distribution, which we don't yet know. For now, however, it will be enough that μ is independent of k; we will derive an expression for its exact value in a moment.

Equations (14.105) and (14.106) can be rearranged to give

$$p_c = \frac{\mu/c}{a_c + \mu/c}$$

(14.107)

and

$$p_k = \frac{a_{k-1}}{a_k + \mu/c}\, p_{k-1}.$$

(14.108)

Applying the latter repeatedly we get

$$\begin{aligned} p_k &= \frac{a_{k-1}a_{k-2}\ldots a_c}{(a_k + \mu/c)\ldots(a_{c+1} + \mu/c)}\, p_c \\ &= \frac{\mu}{ca_k}\, \frac{a_k \ldots a_c}{(a_k + \mu/c)\ldots(a_c + \mu/c)} \\ &= \frac{\mu}{ca_k} \prod_{r=c}^{k} \left[1 + \frac{\mu}{ca_r}\right]^{-1}. \end{aligned}$$

(14.109)

All we need to complete our solution is the value of μ. Taking Eq. (14.102) and letting $n \to \infty$, we get

$$\mu = \sum_{k=c}^{\infty} a_k p_k = \frac{\mu}{c} \sum_{k=c}^{\infty} \prod_{r=c}^{k} \left[1 + \frac{\mu}{ca_r}\right]^{-1}.$$

(14.110)

Canceling μ from both sides we arrive at the equation

$$\sum_{k=c}^{\infty} \prod_{r=c}^{k} \left[1 + \frac{\mu}{ca_r}\right]^{-1} = c.$$

(14.111)

In principle we should be able to solve this equation for μ and substitute the result into Eq. (14.109) to get the complete degree distribution. In practice, unfortunately, the equation is not solvable in closed form for most choices of the attachment kernel a_k, although an approximate value for μ can usually be calculated numerically on a computer. Even without knowing μ, however, we can still find the overall functional form of p_k, which is enough to answer many of the questions we are interested in.

As an example, consider a network of the type observed by Jeong *et al.* [165] and discussed above in which attachment goes as k^γ for some positive constant γ, and let us assume that (as found by Jeong *et al.*) we have $\gamma < 1$. The solution for this particular choice was given by Krapivsky *et al.* [189] and shows a number of interesting features.

Putting $a_k = k^\gamma$ in Eq. (14.109) gives

$$p_k = \frac{\mu}{ck^\gamma} \prod_{r=c}^{k} \left[1 + \frac{\mu}{cr^\gamma}\right]^{-1}.$$

(14.112)

This degree distribution turns out *not* have a power-law tail, by contrast with the case of linear preferential attachment. In other words the power-law form is sensitive to the precise shape of the attachment kernel. We can see this by writing

$$\prod_{r=c}^{k} \left[1 + \frac{\mu}{cr^\gamma}\right]^{-1} = \exp\left[-\sum_{r=c}^{k} \ln\left(1 + \frac{\mu}{cr^\gamma}\right)\right]$$

(14.113)

and then expanding the logarithm as a Taylor series in μ/cr^γ:

$$\begin{aligned} \sum_{r=c}^{k} \ln\left(1 + \frac{\mu}{cr^\gamma}\right) &= -\sum_{r=c}^{k} \sum_{s=1}^{\infty} \frac{(-1)^s}{s} \left(\frac{\mu}{c}\right)^s r^{-s\gamma} \\ &= -\sum_{s=1}^{\infty} \frac{(-1)^s}{s} \left(\frac{\mu}{c}\right)^s \sum_{r=c}^{k} r^{-s\gamma}. \end{aligned}$$

(14.114)

The sum over r cannot be expressed in closed form, but we can approximate it using the

trapezoidal rule, [219] which says that for any function $f(r)$:

$$\sum_{r=a}^{b} f(r) = \int_a^b f(r)\,dr + \tfrac{1}{2}[f(a) + f(b)] + O(f'(b) - f'(a)).$$

(14.115)

(For those not familiar with it, the derivation of the trapezoidal rule is illustrated in Fig. 14.7.[220])

In our case $f(r) = r^{-s\gamma}$ and Eq. (14.115) gives

$$\sum_{r=c}^{k} r^{-s\gamma} = A_s + \frac{k^{1-s\gamma}}{1-s\gamma} + \tfrac{1}{2}k^{-s\gamma} + O(k^{-(s\gamma+1)}),$$

(14.116)

where A_s is a constant depending on s (and on c) but not on k.

Consider now what happens when k becomes large. Since $\gamma > 0$, the term in $k^{-s\gamma}$ and all subsequent terms vanish as $k \to \infty$ and Eq. (14.114) becomes

$$\sum_{r=c}^{k} \ln\left(1 + \frac{\mu}{cr^{\gamma}}\right) \simeq A - \sum_{s=1}^{\infty} \frac{(-1)^s}{s}\left(\frac{\mu}{c}\right)^s \frac{k^{1-s\gamma}}{1-s\gamma},$$

(14.117)

where A is a k-independent constant equal to $\sum_{s=1}^{\infty} A_s(-\mu/c)^s/s$.

This expression can be simplified still further by noting that, in the limit $k \to \infty$, all terms in $k^{1-s\gamma}$ where $1 - s\gamma < 0$ also vanish. Thus for any given value of γ we need keep terms in k up to only a certain value of s. The simplest case is when $\frac{1}{2} < \gamma < 1$. In this case only the term for $s = 1$ grows as k increases, all others vanishing, and

$$\sum_{r=c}^{k} \ln\left(1 + \frac{\mu}{cr^{\gamma}}\right) \simeq A + \frac{\mu k^{1-\gamma}}{c(1-\gamma)}$$

(14.118)

as $k \to \infty$.

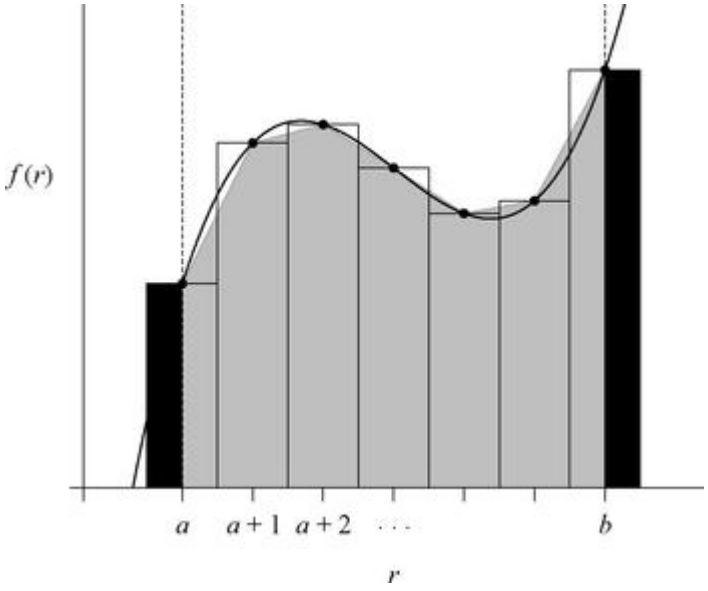


Figure 14.7: The trapezoidal rule. The trapezoidal rule approximates a sum by an integral (or vice versa). The sum of the function $f(r)$ from $r = a$ to $r = b$ (dotted lines) is equal to the sum of the areas of the rectangular bars, which is also equal to the area shaded in gray. This shaded area can be approximated by the integral of f (r) between a and b (smooth curve) plus the two extra rectangular sections at either end (hatched), which have area $\frac{1}{2}f(a)$ and $\frac{1}{2}f(b)$ respectively. Add everything up and we get Eq. (14.115). The error in the approximation is equal to the sum of the relatively small regions between the curve and the shaded area.

Now, combining Eqs. (14.112), (14.113), and (14.118), we find that the asymptotic form of p_k is

$$p_k \sim k^{-\gamma} \exp\left(-\frac{\mu k^{1-\gamma}}{c(1-\gamma)}\right),$$

(14.119)

for $\frac{1}{2} < \gamma < 1$.

Distributions of this general form, in which the dominant contribution to the probability falls off as the exponential of a power of k, are called *stretched exponentials*. Since the exponent $1 - \gamma$ is less than one, the distribution falls off more slowly than an ordinary exponential in k, which is why we called it "stretched."[221] On the other hand, the distribution still falls off a good deal faster than the power law that we found in the case of linear preferential attachment, and this is really the important point here. This calculation reveals that the power-law distribution in the Barabási-Albert model is a special feature of the linear attachment process assumed by that model. (Note that this observation is valid even though we haven't calculated the value of the constant μ. The general functional form of the degree distribution doesn't depend on the value of the constant.)

For other values of γ the calculation is similar but involves more terms in Eq. (14.117). For instance, if $\frac{1}{3} < \gamma < \frac{1}{2}$ then the terms in $k^{1-s\gamma}$ for $s = 1$ and 2 both grow as k becomes large while all others vanish, and we find that

$$\sum_{r=k_0+1}^{k} \ln\left(1+\frac{\mu}{cr^{\gamma}}\right) \simeq A + \frac{\mu k^{1-\gamma}}{c(1-\gamma)} - \frac{\mu^2 k^{1-2\gamma}}{2c^2(1-2\gamma)},$$

(14.120)

which gives

$$p_k \sim k^{-\gamma} \exp\left(-\frac{\mu k^{1-\gamma}}{c(1-\gamma)} + \frac{\mu^2 k^{1-2\gamma}}{2c^2(1-2\gamma)}\right).$$

(14.121)

In between the solutions (14.119) and (14.121) there is a special case solution when γ is exactly equal to $\frac{1}{2}$ .For $\gamma = \frac{1}{2}$ and $s = 2$ the integral in Eq. (14.115) gives rise not to a power of k but to a log and Eq. (14.114) becomes

$$\sum_{r=c}^{k} \ln\left(1+\frac{\mu}{cr^{\gamma}}\right) \simeq A + \frac{2\mu}{c}\sqrt{k} - \frac{\mu^2}{2c^2}\ln k,$$

(14.122)

all other terms vanishing in the limit of large k. Substituting this expression into Eq. (14.113), we then arrive at

$$p_k \sim (\sqrt{k})^{\mu^2/c^2-1} \exp\left(-\frac{2\mu}{c}\sqrt{k}\right),$$

(14.123)

for $\gamma = \frac{1}{2}$.

We can continue in this vein ad infinitum. There are distinct solution forms for $\frac{1}{4} < \gamma < \frac{1}{3}$ and $\frac{1}{5} < \gamma < \frac{1}{4}$ and so forth, as well as special case solutions for $\gamma = \frac{1}{3}, \frac{1}{4}, \frac{1}{5}$, and so forth. Figure 14.8 shows the degree distribution for the case $\gamma = 0.8$, along with the asymptotic form (14.119). Note the convex form of the curve on the semilogarithmic scales, which indicates a function decaying slower than an exponential.

One can also calculate the degree distribution for superlinear preferential attachment, i.e., for values of γ greater than one. This case also shows some nent interesting behaviors: it turns out that for $\gamma > 1$ the typical behavior is for one vertex to emerge as a "leader" in the network, gaining a non-zero fraction of all edges, with the rest of the vertices having small degree (almost all having

degree less than some fixed constant). Readers interested in these developments can find them described in detail in Ref. [189].

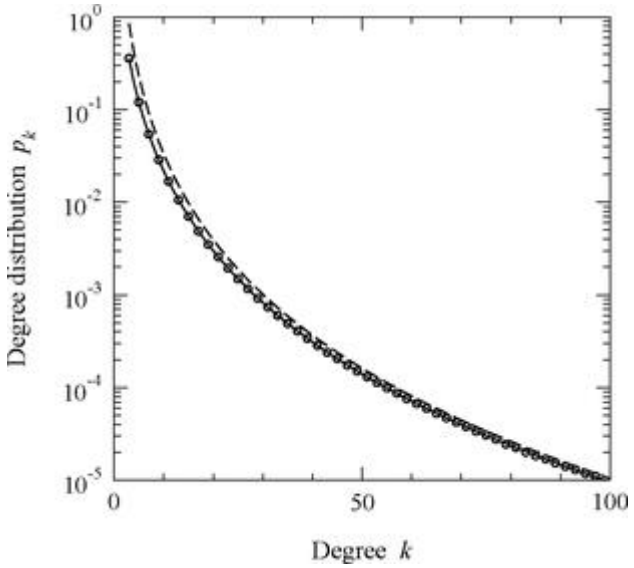


Figure 14.8: Degree distribution for sublinear preferential attachment. This plot shows the fraction p_k of vertices with degree k in a growing network with attachment kernel k^γ as described in the text. In this case $\gamma = 0.8$ and $c = 3$. The points are results from computer simulations, averaged over 100 networks of (final) size 10^7 vertices each. The solid line is the exact solution, Eq. (14.112), evaluated numerically. The dashed line is the asymptotic form, Eq. (14.119), with the overall constant of proportionality chosen to coincide with the exact solution for large values of k.

14.4.4 VERTICES OF VARYING QUALITY OR ATTRACTIVENESS

The models of growing networks we have examined so far assume that all vertices of a given degree are equally likely to gain a new edge. In these models, for example, all papers that have never been cited before are equally likely to get new citations. All websites that no one has linked to yet are equally likely to receive links.

See Sections 7.4 and 19.1 for a discussion of the operation of search engines.

In the real world, of course, nothing could be farther from the truth. There are huge differences in the perceived importance and quality of scientific papers or websites that mean some are far more likely to gain edges than others. A website, for instance, that provides a useful service, such as a directory or an encyclopedia, will almost certainly receive new links at a higher rate than most people's personal home pages. Indeed, search engines use the numbers of links web pages receive precisely as a measure of which pages people find most useful. Similarly, people look at the numbers of citations a paper receives to try to gauge how influential that paper has been. These approaches would not work unless there were some correlation between the degree and the perceived quality of a vertex.

If one allows for variations in the intrinsic quality or attractiveness of vertices, then, presumably it will have an effect on the degree distribution. It seems entirely possible that, with such effects at work, the power laws generated by preferential attachment models might completely disappear, leaving us at a loss to explain how power laws might arise in real-world networks. In this section we study a model of the growth of a network proposed by Bianconi and Barabási [42,43] that includes effects of varying node quality—or *fitness* as they call it. As we will see, the power-law behavior of traditional models disappears once vertex fitness enters the picture, although the distribution for vertices of a given fitness still follows a power law.

The model of Bianconi and Barabási is defined as follows. Vertices are added one by one with each attaching by undirected edges to c prior vertices, just as before. Now, however, each vertex i has a fitness η_i that is assigned at the moment of the vertex's creation and never changed thereafter. The fitnesses are real numbers with values drawn from some distribution $\rho(\eta)$, so that the probability of a value falling between η and $\eta + d\eta$ is $\rho(\eta)\, d\eta$. Each of the c new edges added with each new vertex attaches to a previously existing vertex with probability proportional to an attachment kernel $a_k(\eta)$ that depends now on both the degree k of the target vertex and its fitness η. (In fact, Bianconi and Barabási examined only the special case $a_k(\eta) = \eta k$. The general model considered here was proposed and solved subsequently by Krapivsky and Redner [188].)

This model can be solved by the same method as the model of Section 14.4.3. We define $p_k(\eta, n)\, d\eta$ to be the fraction of vertices with degree k and fitness in the interval η to $\eta + d\eta$ when the network has n vertices. Writing down a master equation as before and taking the limit $n \to \infty$ we arrive at equations that read

$$\dot{p}_k(\eta) = \frac{c}{\mu}\left[a_{k-1}(\eta)p_{k-1}(\eta) - a_k(\eta)p_k(\eta)\right]$$

(14.124)

for $k > c$ and

$$p_c(\eta) = \rho(\eta) - \frac{c a_c(\eta)}{\mu} p_c(\eta),$$

(14.125)

where $p_k(\eta) = p_k(\eta, \infty)$ and μ is again the appropriate normalizing factor

$$\mu = \frac{1}{n}\sum_{i=1}^{n} a_{k_i}(\eta_i) = \sum_{k=c}^{\infty} \int_{-\infty}^{\infty} a_k(\eta) p_k(\eta)\, \mathrm{d}\eta.$$

(14.126)

(See Eq. (14.110).) Note that the +1 of Eq. (14.106) has been replaced by $\rho(\eta)$ in Eq. (14.125), because the average number of new vertices of degree c added to the network with fitness in the interval η to $\eta + \mathrm{d}\eta$ is not 1 but $\rho(\eta)$.

Following the same steps that led to Eq. (14.109), we can solve the master equation to show that

$$p_k(\eta) = \rho(\eta) \frac{\mu}{c a_k(\eta)} \prod_{r=c}^{k} \left[1 + \frac{\mu}{c a_r(\eta)}\right]^{-1},$$

(14.127)

and the value of μ can be determined by substituting this result back into Eq. (14.126) (although usually an analytic solution is not possible and the equations must be solved numerically).

As an example consider the case where the attachment kernel is linear in the degree, $a_k(\eta) = \eta k$, which was studied by Bianconi and Barabási.[222] Since $a_k(\eta)$ is (proportional to) a probability it cannot be negative, so we must restrict η to non-negative values. Then the product in Eq. (14.127) becomes

$$\begin{aligned}
\prod_{r=c}^{k} \left[1 + \frac{\mu}{c\eta k}\right]^{-1} &= \prod_{r=c}^{k} \frac{k}{k + \mu/c\eta} \\
&= \frac{\Gamma(k+1)\Gamma(c + \mu/c\eta)}{\Gamma(c)\Gamma(k + 1 + \mu/c\eta)} \\
&= \frac{c\eta k}{\mu} \frac{\mathrm{B}(k, 1 + \mu/c\eta)}{\mathrm{B}(c, \mu/c\eta)},
\end{aligned}$$

(14.128)

where we have made use of Eq. (14.17) and $\mathrm{B}(x, y)$ is Euler's beta function, Eq. (14.19), again. Substituting into Eq. (14.127), we then find that

$$p_k(\eta) = \rho(\eta)\,\frac{\mathrm{B}(k, 1+\mu/c\eta)}{\mathrm{B}(c, \mu/c\eta)}.$$

(14.129)

We showed previously that the beta function goes as a power law $\mathrm{B}(x, y) \sim x^{-y}$ for large values of its first argument (Eq. (14.25)) so Eq. (14.129) implies that the distribution of the degrees of vertices with a particular value of the fitness η has a power-law tail with exponent

$$\alpha(\eta) = 1 + \frac{\mu}{c\eta}.$$

(14.130)

However, the overall degree distribution for the entire network may or may not have a power-law tail, depending on the distribution $\rho(\eta)$. It is clear that it is a power law for some choices of $\rho(\eta)$—for example the trivial choice where all vertices have the same η, which just reduces to the original Barabási-Albert model. If η is broadly distributed, however, the degree distribution will be a sum over power laws with a wide range of different exponents, which will not in general yield another power law.

The solution above does not tell the whole story. There are some interesting features of this model that are missing from Eq. (14.129). To see this, let us calculate the average degree of a vertex in our network. This might seem like a pointless exercise—the average degree must take the value $2c$ since exactly c edges are added for every vertex—but in fact the calculation is quite revealing. The average degree is given by

$$\begin{aligned}\langle k\rangle &= \sum_{k=c}^{\infty}\int_0^{\infty} k p_k(\eta)\,\mathrm{d}\eta = \sum_{k=c}^{\infty}\int_0^{\infty} k\rho(\eta)\,\frac{\mathrm{B}(k, 1+\mu/c\eta)}{\mathrm{B}(c, \mu/c\eta)}\,\mathrm{d}\eta \\ &= \int_0^{\infty}\frac{\rho(\eta)}{\mathrm{B}(c, \mu/c\eta)}\sum_{k=c}^{\infty} k\,\mathrm{B}(k, 1+\mu/c\eta)\,\mathrm{d}\eta.\end{aligned}$$

(14.131)

The sum can be performed by making use of the integral form of the beta function, Eq. (14.33), and gives

$$\sum_{k=c}^{\infty} k\,\mathrm{B}(k, 1+\mu/c\eta) = \frac{c}{1-c\eta/\mu}\,\mathrm{B}(c, \mu/c\eta).$$

(14.132)

An important point to notice, however, is that this result only works if $\eta < \mu/c$. If $\eta \geq \mu/c$ the sum diverges making the average degree in the network infinite, which cannot be the case since, as we have said, the average degree is always $2c$. To avoid the divergence we will impose the restriction that $\rho(\eta) = 0$ for all $\eta \geq \eta_0$, where η_0 is a constant in the range $0 \leq \eta_0 < \mu/c$. (The interesting question of what happens to the network if we choose a $\rho(\eta)$ that violates this condition is dealt with below.)

Combining Eqs. (14.131) and (14.132), we then find that

$$\langle k \rangle = c \int_0^{\eta_0} \frac{\rho(\eta)\,d\eta}{1 - c\eta/\mu}.$$

(14.133)

And since $\langle k \rangle = 2c$ this immediately implies that

$$\int_0^{\eta_0} \frac{\rho(\eta)\,d\eta}{1 - c\eta/\mu} = 2.$$

(14.134)

We haven't yet calculated a value for the constant μ, but even without it this equation tells us something interesting. The integral is a monotonically decreasing function of μ: it takes its smallest value of 1 when $\mu \to \infty$, and its largest value when $\mu \to c\eta_0$. (Recall that $\eta_0 < \mu/c$ so μ can get no smaller than $c\eta_0$.) But if this largest value is still less than two then there is no way to satisfy Eq. (14.134) and no value of μ such that Eq. (14.133) gives the correct answer for $\langle k \rangle$. In the limit $\mu \to c\eta_0$ the denominator of the integrand equals $1 - \eta/\eta_0$, which tends to zero at the upper limit of the integral, but provided $\rho(\eta)$ also tends to zero in this limit the integral can take a finite value and, as we will see in a moment, this value can certainly be less than two for some choices of $\rho(\eta)$.

How can it be that our solution does not give the correct value for the average degree? Have we made a mistake somewhere? More importantly, what does the network actually do in this regime?

The answer to this conundrum turns out to be a subtle and interesting one. The important point is that there are some behaviors of the vertex degrees in a growing network that cannot be captured by a simple probability distribution p_k. In particular, if there are a fixed, finite number of vertices in the network with degrees that scale in proportion to the size n of the entire network, those vertices do not appear in the degree distribution: because there are only a fixed number of them they constitute a fraction $1/n$ of the network and hence contribute zero to the degree distribution as $n \to \infty$. Nonetheless, they make a non-zero contribution to the average degree of the network in the limit of large n and hence must be taken into account in the calculation of $\langle k \rangle$.

Bianconi and Barabási referred to the appearance of such vertices in the network as

"condensation" by analogy with similar behaviors seen in lowtemperature physics,[224] and to the vertices themselves as a *condensate*. For some choices of $\rho(\eta)$ this kind of condensation does indeed occur and a condensate of "superhubs" with very high degree forms in the network.

Suppose we are in such a regime and let us write the sum of the degrees of the vertices in the condensate as K. Then the full expression for the average degree, including the condensate, becomes

$$\langle k \rangle = \frac{K}{n} + \sum_{k=c}^{\infty} \int_0^{\infty} k p_k(\eta)\, d\eta = \frac{K}{n} + c \int_0^{\eta_0} \frac{\rho(\eta)\, d\eta}{1 - c\eta/\mu}.$$

(14.135)

Clearly, no matter what the value of the integral, it is now always possible to achieve $\langle k \rangle = 2c$ by making K sufficiently large. The appropriate value of K is given by setting $\langle k \rangle = 2c$ and rearranging thus:

$$K = nc \left[2 - \int_0^{\eta_0} \frac{\rho(\eta)\, d\eta}{1 - c\eta/\mu} \right].$$

(14.136)

Again, the largest value of the integral is achieved when $\mu \to c\eta_0$, which means that

$$K > nc \left[2 - \int_0^{\eta_0} \frac{\rho(\eta)\, d\eta}{1 - \eta/\eta_0} \right].$$

(14.137)

So if the value of this integral is less than two, then in order to get the correct value for the average degree we require that K scales with the size n of the network just as we hypothesized.

As an example, suppose the distribution of fitnesses takes the form $\rho(\eta) = A(\eta_0 - \eta)^{\tau}$, where τ is a positive exponent and A is a normalization constant τ+1 whose value is easily shown to be $A = (\tau + 1)/\eta_0^{\tau+1}$. Then the integral in Eq. (14.137) is

$$\int_0^{\eta_0} \frac{\rho(\eta)\, d\eta}{1 - \eta/\eta_0} = \frac{\tau + 1}{\eta_0^{\tau}} \int_0^{\eta_0} (\eta_0 - \eta)^{\tau - 1}\, d\eta = 1 + \frac{1}{\tau},$$

(14.138)

and

$$K > nc\left(1 - \frac{1}{\tau}\right).$$

(14.139)

If $\tau < 1$ then this result tells us only that $K \geq 0$, which is trivially true. But if $\tau > 1$ then the sum of the degrees in the condensate must vary in proportion to the network size n.

Thus, depending on our choice for the distribution of fitnesses, the network can show two different behaviors. In one case, the distributions of the degrees of vertices with any given fitness follow a power law with a fitness-dependent exponent, but no vertices in the network are special or distinguished by any particular behavior. In the other case, a condensate forms consisting of one or more "superhubs," which connect to a non-zero fraction of all other vertices. None of the remaining non-condensate vertices show any special behavior, however; they still have power-law degree distributions for each value of fitness, with the same exponents as before. Some authors have likened the condensation phase of the model to the monopolistic dominance of a market by a single vendor or a small number of vendors—once one vertex (vendor) gets a non-zero fraction of all edges (business), preferential attachment guarantees that it will go on doing so thereafter.

We still do not have a complete solution of the model, because we are missing the value of μ which is required to evaluate Eq. (14.136). Unfortunately, to calculate μ we need to know the exact form of the condensate. In most discussions of the model in the literature it is assumed that the condensate consists of just a single vertex of degree K at or close to the maximum fitness η_0, in which case we can evaluate μ using Eq. (14.126). Including the contribution from the condensate vertex, this equation becomes

$$\mu = \frac{a_K(\eta_0)}{n} + \sum_{k=c}^{\infty} \int_{-\infty}^{\infty} a_k(\eta) p_k(\eta)\, d\eta,$$

(14.140)

and, setting $a_k(\eta) = \eta k$ again and making use of Eqs. (14.129) and (14.132), we then find

$$\frac{\eta_0 K}{n\mu} + \int_0^{\eta_0} \frac{\rho(\eta)\, d\eta}{\mu/c\eta - 1} = 1.$$

(14.141)

Equations (14.136) and (14.141) together now give us two equations in the two unknowns K and μ, which we can, at least in theory, solve for both given $\rho(\eta)$, although in practice closed-form solutions are rare because the integrals are non-trivial.

There is, however, no reason why the condensate must consist of just a single vertex. It could in principle consist of more than one. It could even consist not of a fixed number of vertices but of a growing number provided the number grows slower than linearly with the size of the network, so that again the condensate makes no contribution to the degree distribution p_k in the limit of large n. In the latter case, the superhubs that make up the condensate would have degrees that themselves scaled sublinearly with system size, but would still become arbitrarily large as $n \to \infty$. To the best of the author's knowledge, it is not known how to predict which of these behaviors will happen for a given choice of $\rho(\eta)$. Indeed, an exact prediction may not even be possible: computer simulations of the model appear to indicate that the exact nature of the condensate—how many vertices it contains and how their degrees grow with system size—is not deterministic but depends on the details of fluctuations taking place in the early growth of the network. If one performs repeated computer simulations with the same choice of $\rho(\eta)$, the macroscopic behavior of the condensate varies from one run of the program to another.[225]

We have also assumed in our discussion that the distribution of the fitness is bounded, that there is a maximum value η_0 that the fitness can take. What happens if this assumption is violated? In this case there will of course still be a fittest vertex in the network and the network cannot "tell" whether the fitness distribution is bounded above that point or extends to $\eta = \infty$, and hence the behavior of the model will be the essentially the same as in the bounded case. The main difference from the bounded case is that the value of the highest fitness may change from time to time, which also changes the value of μ via Eq. (14.126). However, the changes in the highest fitness become rarer and rarer as time goes by[226] so that asymptotically the behavior of the system is the same in the bounded and unbounded cases for arbitrarily long periods of time.

Many other extensions and variations of preferential attachment models have been studied in addition to the ones described in this chapter. If you're interested in learning more, there are a number of review articles that go into the subject in some detail—see Refs. [12], [46], and [98]. The rest of this chapter is devoted to the discussion of other models of network formation and growth that don't rely on preferential attachment.

14.5 VERTEX COPYING MODELS

Preferential attachment models offer a plausible, if simplified, explanation for power-law degree distributions in networks such as citation networks and the World Wide Web. Preferential attachment, however, is by no means the only mechanism by which a network can grow, nor even the only mechanism known to generate power laws. In the remainder of this chapter we look at a number of other models and mechanisms for the formation of networks, starting in this section with models based on vertex copying.

In Section 14.1 we introduced the preferential attachment mechanism and suggested a possible explanation of its origin in citation networks, that a reader perusing the literature in a given academic field would encounter citations to frequently cited papers more often than citations to less cited ones, and hence would be more likely to cite those frequently cited papers themselves. Another way of saying this is that, in effect, researchers are copying citations from the bibliographies of papers they read.[227]

Kleinberg *et al.* [180] have proposed an alternative mechanism for network formation that takes this idea one step further. What if people simply copied the entire bibliography of a single paper to create the new bibliography of their own paper? This would then create a new vertex in the network with the same pattern of outgoing edges as the vertex they copied from.

As we will see, this process, with slight modifications, can give rise to a power-law degree distribution. First, however, we note that the process as stated has some problems. To begin with, it's clearly rather far-fetched. Authors of papers do take note of who other authors have cited, but it seems unlikely that an author would copy the entire bibliography from someone else's paper. Moreover, if they did just copy the entire bibliography then previously cited papers would get new citations as a result, but there would be no way for papers to receive citations if they had never been cited before.

Both of these problems can be solved by changing the model a little. Instead of assuming that the bibliography of the new paper is copied wholesale from the bibliography of an older one, let us assume that only some fraction of the entries in the old bibliography are copied. Then the remainder of the new bibliography is filled out with references to other papers. These other papers could be selected in a variety of way, but a simple choice would be to select them uniformly at random from the entire network.

These modifications insure that bibliographies are now no longer copied in their entirety and papers with no previous citations have a chance of being cited. The model is, however, still not a very plausible model of a real citation network. But, like Price's preferential attachment model (which is also not very realistic), it can be regarded as a simplified and tractable version of the vertex copying mechanism that allows us to investigate quantitatively the consequences of that mechanism.[228] The precise definition of the model is as follows.

Let us suppose for simplicity that each new vertex added to our network has the same out-degree c. In the language of citations, the bibliographies are all the same size. For each vertex added we choose uniformly at random a previous vertex and go one by one through the c entries in the bibliography of that previous vertex. For each entry we either (a) with probability $\gamma < 1$ copy that entry to the bibliography of the new vertex or (b) with probability $1 - \gamma$ add to the bibliography of the new vertex a citation to another vertex chosen uniformly at random from the entire network. The end result is a bibliography for the new vertex in which, on average, γc of the entries are copied from the old vertex and the remainder are chosen at random. In effect, we have made an imperfect copy of the old vertex in which the destinations of some fraction of the outgoing edges have been randomly reassigned.

We also need to specify the starting state of the network, but, as with our preferential attachment

models, it turns out that the asymptotic properties of the network do not depend on the state we choose. Thus the choice is not particularly important, but we could, for instance, specify a starting network consisting of some number $n_0 > c$ vertices in which each points randomly to c of the others.

We can solve for the degree distribution of the network generated by this model as follows. Let us ask what the probability is that vertex i receives a new incoming edge upon the addition of a new vertex to our network. For i to receive a new edge, one of two things has to happen. Either the newly added vertex happens to copy connections from a vertex that already points to vertex i, in which case with probability γ the connection to i will itself get copied, or i could be one of the vertices chosen at random to receive a new edge. Let us treat these two processes separately.

Suppose that a particular existing vertex happens to have a link to our vertex i. The probability that a newly added vertex will choose to copy its own links from this existing vertex is simply $1/n$, since the source for the copies is chosen uniformly at random from the whole network. Thus if i has in-degree q_i, the chance that any one of the q_i vertices that point to it gets chosen is q_i/n. And the chance that the link from that vertex to i gets copied is γ, for a total probability of $\gamma q_i/n$.

The average number of random links that a newly added vertex makes—ones not copied from a previous vertex—is $1 - \gamma$ for each of its c outgoing edges, or $(1 - \gamma)c$ overall. And the probability that our vertex i happens to be the target of one of these random links is $1/n$, for an overall probability of $(1—\gamma)c/n$.

Putting everything together, the total probability that vertex i gets a new link is

$$\frac{\gamma q_i}{n} + \frac{(1-\gamma)c}{n} = \frac{\gamma q_i + (1-\gamma)c}{n}.$$

(14.142)

Defining $p_q(n)$ as before to be the fraction of vertices with in-degree q when the network has n vertices, the total expected number of vertices of in-degree q receiving a new edge is

$$np_q(n) \times \frac{\gamma q + (1-\gamma)c}{n} = [\gamma q + (1-\gamma)c]\,p_q(n).$$

(14.143)

But now we notice a remarkable fact. If we define a new constant a by

$$a = c\left(\frac{1}{\gamma} - 1\right),$$

(14.144)

then

$$\gamma = \frac{c}{c+a}$$

(14.145)

and Eq. (14.143) becomes

$$[\gamma q + (1-\gamma)c]\,p_q(n) = \frac{c(q+a)}{c+a}\,p_q(n),$$

(14.146)

which is exactly the same as the probability (14.2) for the equivalent quantity in Price's model.

We can now use this probability to write down a master equation for the evolution of the degree distribution p_q, which will be precisely the same as the master equation (14.5) for Price's model and all subsequent developments follow through just as in Section 14.1. The end result is that our vertex copying model behaves precisely as the Price model does, but with a value of a specified now in terms of the parameter γ by Eq. (14.144). Thus, for example, the degree distribution in the limit of large n will obey Eq. (14.21) and hence will asymptotically follow a power law with exponent α given by Eq. (14.27) to be

$$\alpha = 2 + \frac{a}{c} = 1 + \frac{1}{\gamma}$$

(14.147)

This gives exponents in the range from 2 to ∞, with the value depending on how faithfully vertices are copied. Faithful copies (γ close to one) give exponents close to two, while sloppy copies give exponents that can be arbitrarily large. Other properties of Price's model carry over as well, such as the distribution of in-degree as a function of age given in Eq. (14.57).

This is not to say, however, that vertex copying generates networks identical in every respect to preferential attachment. With vertex copying, for instance, many of the links that a newly appearing vertex makes are typically copied from the same other vertex and hence most vertices in the network will have connections that are similar to those of at least one other vertex. In preferential attachment models, on the other hand, there is no such correlation between the connections of different vertices—each link is chosen independently from the available possibilities at the time it is created and not copied from anywhere else. The two networks therefore, while they may have the same degree distribution, are different in the details of their structure.

In addition to being interesting in its own right, the vertex copying model serves as a useful cautionary tale concerning the mechanisms of network formation. We have seen that many real networks have degree distributions that follow a power law, at least approximately, and that preferential attachment models can generate such degree distributions. A natural conclusion is that real networks are the product of preferential attachment processes, and this may indeed be correct.

We should be careful, however, not to jump immediately to conclusions because, as we have now seen, there exists at least one other mechanism—vertex copying—that produces precisely the same degree distribution. Without further information we have no way of telling which of these mechanisms is the correct one, or whether some other third mechanism that we have not yet thought of is at work.

One could in principle examine details of the structure of specific real-world networks in an attempt to tell which, if either, of our two mechanisms is the better model for their creation. For instance, one might examine a network to see if there appear to be pairs of vertices whose outgoing connections are approximate copies of one another. In fact, in real citation networks it turns out that there are many such pairs, an observation that appears to lend weight to the vertex copying scenario. However, we must remember that both of our models are much simplified and it's likely that neither of them is an accurate representation of the way real networks are created. A simple explanation for vertices in citation networks with similar patterns of links is that they correspond to papers on similar topics and so tend to cite the same literature; there is no need to assume that one of them copied from the other. As a result, it may not be possible to distinguish firmly between preferential attachment and vertex copying in many cases.

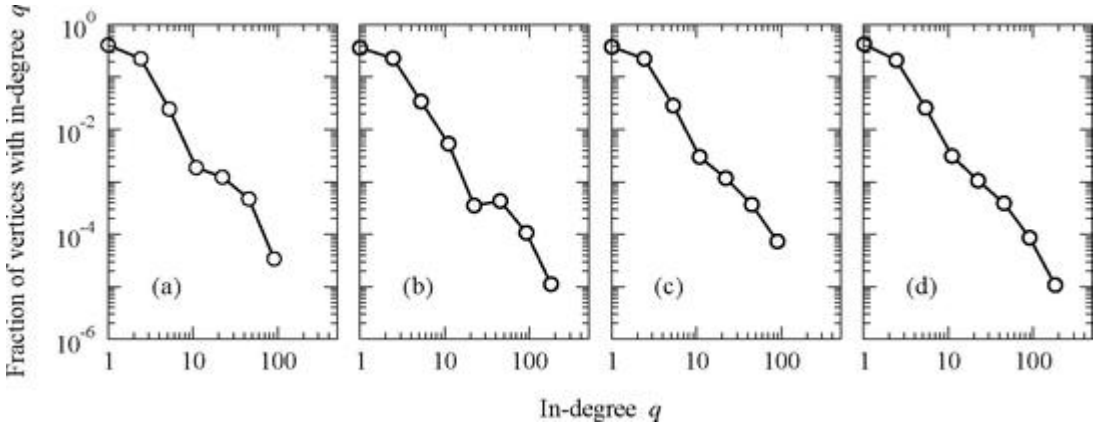


Figure 14.9: Distribution of in-degrees in the metabolic networks of various organisms. Jeong *et al.* [166] examined the degree distributions of the known portions of the metabolic networks of 43 organisms, finding some of them to follow power laws, at least approximately. Show here are the in-degree distributions for (a) the archaeon *A. fulgidus*, (b) the bacterium *E. coli*, (c) the worm *C. elegans* (a eukaryote), and (d) the aggregated in-degree distribution for all 43 organisms. After Jeong *et al.* [166].

There are, however, some cases where preferential attachment appears to be an implausible candidate to explain the structure of a network, and in some of these cases vertex copying is the most promising remaining option. A good example comes from the realm of biology, where vertex copying is considered a strong candidate for explaining the structure of metabolic networks and protein-protein interaction networks. As discussed in Chapter 5, these are networks of chemical and physical interactions between molecules in the cell and, although our knowledge of their structure is currently quite incomplete, there is at least tentative evidence to suggest that they have power-law degree distributions—see Fig. 14.9 and Refs. [164] and [166]. It seems unlikely, however, that preferential attachment is the cause of these power laws: there is no obvious mechanism by which preferential attachment could take place in this context. Vertex copying, on the other hand, may be a reasonable candidate.

Consider for example a protein interaction network. As described in Section 5.1.3, proteins in the cell are created by the processes of molecular transcription and translation from codes stored in the cell's DNA. The section of code that defines a single protein is called a gene and it turns out

that genes are sometimes inadvertently copied when cells reproduce.

When a cell splits in two to reproduce, its DNA is copied so that each half of the split cell will have a complete copy. The cellular machinery responsible for the copying is highly reliable, but not perfect. Very occasionally, a section of DNA will be copied twice, giving rise to a repeated section, which can mean that the new cell has two copies of a certain gene or genes where the old cell had only one. Many examples of such repeated sections are known in the human genome and the genomes of other animals and plants.

Another common type of copying error is the *point mutation*, whereby individual nucleotides—letters in the DNA code—are copied incorrectly. Over the course of many cell divisions, point mutations can accumulate, and as a result two initially identical versions of the same gene can become no longer identical, with some fraction of their bases changed to new and (roughly speaking) random values. These processes typically happen slowly over the course of evolutionary time, taking thousands or even millions of years. The end result, however, is that a gene is copied and then mutated to be slightly different from the original.

And these processes are reflected in the network of protein interactions. Typically both copies of a duplicated gene in a genome can generate the corresponding protein; the subsequent mutation of one or both copies can result in the two producing similar but slightly different versions of the protein, different enough in some cases to also have slightly different sets of interactions in the network. Some interactions may be common to both proteins but, just as in our vertex copying model, some may also be different.

This picture is made more plausible by the fact that changes in genes are not purely random but are subject to Darwinian selection under which some gene mutations are more advantageous than others. A cell with two copies of a particular protein may gain a selective advantage if those copies do slightly different things, rather than needlessly duplicating functionality that a single copy alone could achieve. Thus it seems possible that nature may actually favor duplicated proteins that have slightly different functions and hence different sets of network connections. Moreover an examination of the data for real-world protein-protein interaction networks turns up many examples of pairs of proteins that are similar but not identical in their patterns of interactions, and gene duplication is widely, if not universally, believed to be the cause.

Several models of vertex copying and mutation in biological networks have been proposed and studied. The model proposed by Solé *et al.* [302], for example, is very similar to the model described above, the main difference being that it is a model of an undirected network rather than a directed one. Another model, put forward by Vázquez *et al.* [317], is also similar but includes a mechanism whereby the connections of the *copied* vertex can be changed as well as those of the copying vertex. Although the latter mechanism would make little sense in a model of a citation network (the bibliography of a paper never changes after publication), it is appropriate in the biological context, where all genes are potentially mutating all the time.

14.6 NETWORK OPTIMIZATION MODELS

In the models we've looked at so far in this chapter, network structure is determined by the way in which the network grows—how newly added vertices connect to others, where newly added edges get placed, and so forth. Furthermore, the structure of these networks is for the most part a result of a succession of random processes, often decentralized and quite blind to the large-scale structure they are creating.

An alternative network formation mechanism, important in certain types of network, is structural optimization. In some cases, such as transportation networks (Section 2.4) or distribution networks (Section 2.5), a network has been specifically designed to achieve a particular goal or goals, such as the delivery of mail or packages around the country or the transportation of airline passengers to their destinations, and the structure of the network can heavily influence the efficiency with which that goal is accomplished. Networks of airline routes, for example, are typically based on a hub-and-spoke arrangement with a small number of busy airport hubs and a large number of minor destinations. [230] (Package delivery companies also use a similar scheme.) The reason is that it makes little sense to fly airplanes directly between minor destinations—there will typically be very few passengers interested in the service and the planes will be half empty. By ensuring that the only flights in and out of minor destinations are to and from major hubs, one concentrates the passengers on those routes, ensuring fuller planes while still giving the passengers a reasonably short journey.

In other words, the hub-and-spoke design of the airline networks *optimizes* the network, making it more efficient, and hence more profitable, for the airline. In such cases, the structure of the network is explained not by a growth mechanism but by the fact that the network has been designed to optimize certain characteristics. In this section we look briefly at some models of network optimization.

14.6.1 TRADE-OFFS BETWEEN TRAVEL TIME AND COST

The example given above of an airline network is a good place for us to start. Airline networks are, in fact, highly optimized: the airline industry operates on very small (sometimes even negative) profit margins, and optimization of operations to trim even a tiny percentage off their enormous costs can make a substantial difference to the bottom line. Airlines employ large staffs of researchers whose sole task is to find new ways to optimize aspects of their business, including particularly their network of routes. At the same time, airlines need to keep their customers happy if they are to avoid losing market share to their competitors. This means, for instance, that they need to provide short, quick routes between as many pairs of destinations as possible—travelers are strongly averse to long journeys that wear them out or waste their time. The twin goals of cost-efficient operation and short routes are to some extent at odds with one another. The quickest way to get passengers from any place to any other, for example, would be to fly separate planes between every pair of airports in the country, but this would be immensely costly. The observed structure of real airline networks is a compromise response to the conflicting needs of the company and its passengers.

The optimization problems faced by real airlines are, inevitably, hugely complex, involving as they do organizations with thousands of employees, billions of dollars worth of material resources, and rapidly changing parameters such as fuel costs, consumer demand, and the nature of the competition. Nonetheless, there is insight to be gained by creating and studying simplified models of the optimization process in the same way that simple models of, for example, citation networks can grant us insight despite the many features of real citation processes that they omit.

One of the simplest models of network optimization is that proposed by Ferrer i Cancho and Solé [117], which balances two elements of exactly the types discussed above. In this model the cost of maintaining and operating the network is represented by the number of edges m in the network. This would be equivalent to saying that the cost of running an airline is proportional to the number of routes it operates. Obviously this is a vast simplification of the real situation, but let us accept it for the moment and see where it leads. The customer satisfaction half of the equation is represented by the mean geodesic distance ℓ between vertex pairs. In our airline example ℓ would be the average number of legs required to journey from one point to another, which is certainly one element of customer satisfaction, though not the only one. Technically, ℓ is a *dissatisfaction* measure, since large values correspond to disgruntled customers.

We would like to design a network that minimizes both m and ℓ but this is in general not possible: the minimum value of ℓ is achieved by placing an edge between every pair of vertices, but this maximizes the value of m. Thus our two goals are, as discussed above, at odds with one another and the best we can hope for is a reasonable compromise between them. In search of such a compromise, Ferrer i Cancho and Solé studied the quality function

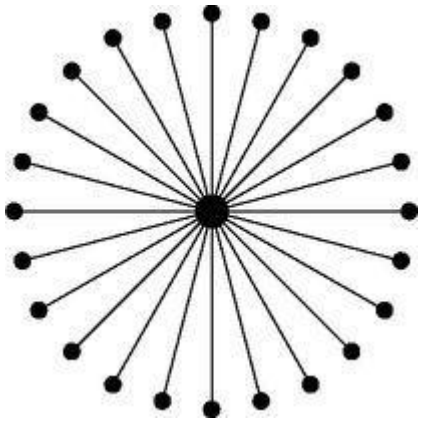
$$E(m,\ell) = \lambda m + (1-\lambda)\ell,$$

(14.148)

where λ is a parameter in the range $0 \le \lambda \le 1$. For any given network and a given value of λ we can calculate $E(m, \ell)$; the value of ℓ for instance can be computed using the breadth-first search algorithm of Section 10.3. Ferrer i Cancho and Solé considered networks of a given number of vertices n and then asked what happens when we try to minimize $E(m, \ell)$ by varying the position

of the edges in that network to find the smallest value possible. If $\lambda = 1$, then $E = m$ and this process is equivalent to just minimizing the number of edges without regard for path lengths. If $\lambda = 0$ then $E = \ell$ and we are minimizing only average path length without regard for m. For values in between, we are striking a balance between number of edges and path length, with the precise weight of each term controlled by our choice of λ.

At some level, this model is a trivial one. Observe that the value of ℓ becomes formally infinite if there is any pair of vertices in the network that is not connected by a path—i.e., if the network has more than one component—since the distance between such pairs is by convention considered infinite (see Section 6.10.1). Thus the minimum value of E must be for a connected network, a network with just one component. Observe also that the minimum value of m for a connected network is $m = n - 1$, where n is the number of edges. This is the value for a tree, which is the connected network with the smallest number of edges (see Section 6.7).



A star graph of 25 vertices.

Provided λ is reasonably large, so that we place a moderate amount of weight on minimizing m, the network with the best value of $E(m, \ell)$ is then found by giving m its minimum value of $n - 1$, which constrains the network to be a tree, and searching through the set of possible trees to find the one that minimizes ℓ. In fact, the latter task has a simple, known solution: the minimum value of ℓ among trees with n vertices is obtained by the *star graph*, the network in which there is a single central hub connected by a single edge to each of the $n - 1$ remaining vertices. By definition there are always exactly m pairs of vertices with geodesic distance one in any network—the pairs that are directly connected by an edge—which means that in a tree there are $n - 1$ such pairs. Among the set of all trees, therefore, the value of the mean distance ℓ is governed by the numbers of pairs with distances of two or more, since the number with distance one is fixed. But in the star graph all other pairs have distance exactly two—the shortest (indeed only) path from any (non-hub) vertex to any other is the path of length two via the hub. Thus there can be no other tree with a smaller value of ℓ.

Thus, for sufficiently large λ, the optimum network under the quality function (14.148) is always the star graph. This is satisfying to some extent: it offers a simple explanation of why the hub-and-spoke system is so efficient. It offers short journeys while still being economic in terms of the number of different routes the airline has to operate. But it is also, as we have said, somewhat trivial. The model shows essentially only the one behavior. For smaller values of λ other behaviors are possible, but it turns out that the value of λ has to be *really* small: non-star-graph solutions only appear when

$$\lambda < \frac{2}{n^2 + 2}.$$

(14.149)

Since the expression on the right-hand side dwindles rapidly as n becomes large, the optimal network is the star graph for almost all values of λ, even for networks of quite modest size.

In their paper, however, Ferrer i Cancho and Solé did not perform precisely the calculation we have done here. Instead, they took a different and interesting approach, in which they looked for *local minima* of $E(m, \ell)$, rather than the global minimum. They did this numerically, starting with a random network, repeatedly choosing a pair of vertices at random, and either connecting them by an edge if they were not already connected or deleting the edge between them if they were. Then they compared the value of E before and after the change. If E decreased or stayed the same, they kept the change. If not, they reverted back to the state of the network before the change. The whole procedure was then repeated until the value of E stopped improving, meaning in practice that a long string of attempted changes were rejected because they increased E.

An algorithm of this kind is called a *random hill climber* or *greedy algorithm*. The networks it finds are networks for which the value of E cannot be reduced any further by the addition or removal of any single edge. This does not mean, however, that no lower values of E exist: there may be states of the network that differ by more than one edge—the addition and deletion of whole regions of the network—that have better values of E. But if so, the algorithm will not find them. It comes to a halt at a local minimum where no single-edge change will improve the value of E.

When studied in this way, the model shows an interesting behavior. For large values of λ, where the addition of an edge costs a great deal in terms of the value of E, the algorithm rapidly runs into trouble and cannot find a way to improve the network, long before it gets anywhere close to the optimum hub-and-spoke arrangement. When λ is small, on the other hand, the algorithm typically manages to find the star graph solution. The result is a spectrum of networks that range from a random-looking tree to a star-graph, as shown in Fig. 14.10.

What's more, Ferrer i Cancho and Solé found that the degree distributions of their networks show interesting behavior, passing from an exponential distribution for large λ, though a transition point with a power-law degree distribution, to approximately star-like graphs for small λ in which one vertex gets a finite fraction of all the edges and the remaining vertices have low degree. This spectrum is reminiscent of the behavior of continuous phase transitions such as the transition at which a giant component appears in a random graph (see Section 12.5), in which an initially exponential distribution of component sizes passes through a transition to a regime in which one component gets a finite fraction of all vertices and the rest are small.

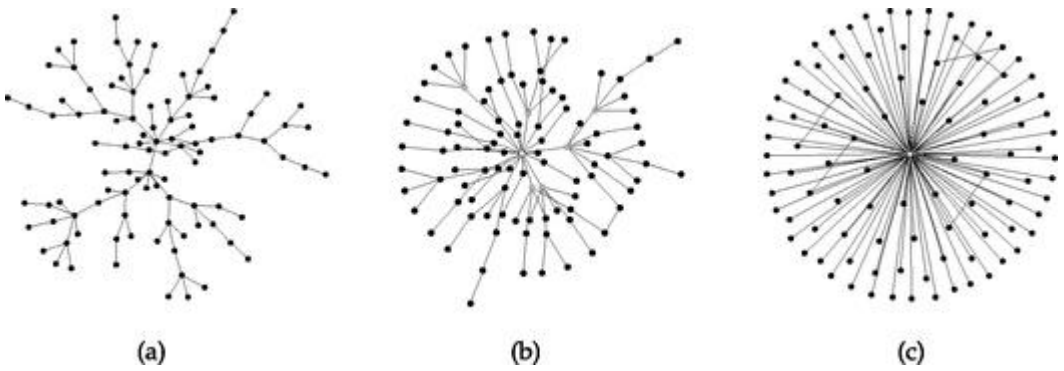


Figure 14.10: Networks generated by the optimization model of Ferrer i Cancho and Solé. The optimization model described in the text generates a range of networks, all trees or

approximate trees, running from (a) distributed networks with exponential degree distributions, through (b) power-law degree distributions, to (c) star graphs in which there is just one major hub. Figure adapted from [117]. Original figure Copyright 2003 Springer-Verlag Berlin Heidelberg. Reproduced with kind permission of Springer Science and Business Media.

Sadly, this observation does not go any further than an intriguing hint. The work of Ferrer i Cancho and Solé is entirely numerical and they do not give any analytic treatment of the model. In addition there are some other problems with the model. In particular, it is not clear why one should look at local minima of E rather than global ones: the researchers who work for real airlines are certainly capable of realizing when they are stuck in a local optimum and better profits are available by changing the network in some substantial way that moves them to a different and better optimum. It seems likely therefore that, to the extent that real networks show interesting structural behavior of the type observed here, it is not a result of getting stuck in local minima and hence that a model with a different approach is needed.

One such model, proposed by Gastner and Newman [137], generalizes that of Ferrer i Cancho and Solé by considering not only number of legs in a journey but also the geographic distance traveled. Suppose that airline travelers are principally concerned not with the number of legs in their journey but with the total time it takes them to travel from origin to destination. Number of legs can be regarded as a simple proxy for travel time, but a better proxy would be to take the length of those legs into account as well as their number. The travel time contributed to a journey by one leg is composed of the time spent in the airport (checking in, waiting, embarking, taxiing, disembarking, etc.) plus the time spent in the air. A simple formula would be to assume that the former is roughly constant, regardless of the distance being traveled, while the latter is roughly proportional to the distance traveled. Thus, the time taken by a leg from vertex i to vertex j in our network would be

$$t_{ij} = \mu + v r_{ij},$$

(14.150)

where μ and v are constants and r_{ij} is the distance flown from i to j. By varying the values of μ and v, we can place more or less emphasis on the fixed "airport" time cost versus the time spent in the air.

Gastner and Newman used this expression for travel time in place of the simple hop-count of the model of Ferrer i Cancho and Solé, redefining ℓ to be the average shortest-path distance between pairs of vertices when distances are measured in terms of travel time. The quality function E is defined just as before, Eq. (14.148), but using this new definition of ℓ.

Despite the superficial similarity between this model and that of Ferrer i Cancho and Solé there is a crucial difference between the two: the model of Gastner and Newman depends on actual spatial distances between airports and hence requires that the vertices of the network be placed at some set of positions on a map. The model of Ferrer i Cancho and Solé by contrast depends only on the network topology and has no spatial element. There are various ways in which the vertices can be positioned on the map. Gastner and Newman, for instance, specifically considered the map of the United States and took the real US population distribution into account, placing vertices with greater density in areas with greater populations. While this adds a level of realism to the calculations, the interesting behavior of their model can be seen without going so far. In the examples given here we consider a fictional map in which vertices are just placed uniformly at random in a square with periodic boundary conditions.

Another important difference between the two models is that Gastner and Newman considered the global optimum of the quality function rather than local optima as Ferrer i Cancho and Solé

did. In practice, unfortunately, the global optimum is hard to find, so one often has to make do with approximate optima. Gastner and Newman used the numerical optimization technique called simulated annealing to find good approximations to the global optimum, but we should bear in mind that they are only approximations.

Figure 14.11 shows optimal or approximately optimal networks for various values of the parameters μ and v. The leftmost frames of the figure correspond to small μ and large v, meaning that the cost to the traveler of a trip is roughly proportional to the total mileage traveled and the number of legs has little effect. In this case, the best networks are ones that allow travelers to travel in roughly straight lines from any origin to any destination. As the figure shows, the networks are roughly planar in appearance. They look reminiscent of road networks, rather than airline routes, and this is no coincidence. Travel times for road travelers are indeed dominated by total mileage: there is almost no ″per leg″ cost associated with road travel, since it takes only a few seconds to turn from one road onto another. It is satisfying to see therefore that the simple model of Gastner and Newman generates networks that look rather like real road maps in this limit.

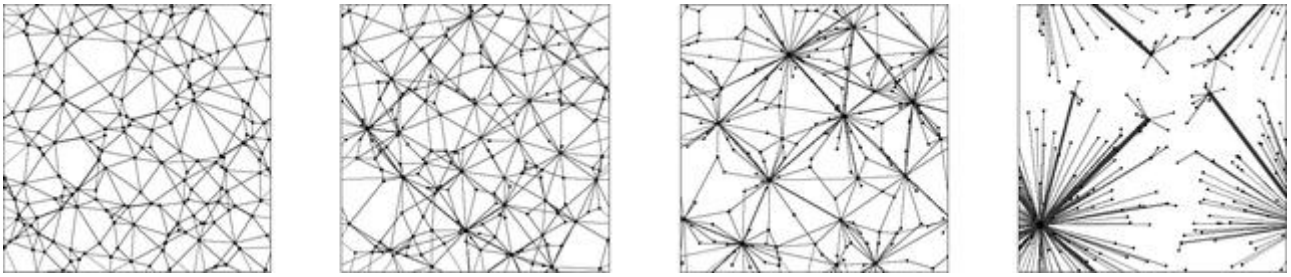


Figure 14.11: Networks generated by the spatial network model of Gastner and Newman. The four frames show networks that optimize or nearly optimize the quality function, Eq. (14.148), with ℓ defined according to the prescription of Gastner and Newman [137] in which the lengths of edges in the network are chosen to represent the approximate travel time to traverse the edge. Travel time has two components, a fixed cost per edge and a cost that increases with the Euclidean length of an edge. The frames show the resulting networks as the relative weight of these two components is varied between the extremes represented by the network on the left, for which all of the weight is on the Euclidean length, and by the network on the right, for which cost is the same for all edges. The resulting structures range from road-like in the former case, to airline-like in the latter. Adapted from Gastner [135].

The rightmost frames in the figure show optimal networks for large μ and small v—the case where it is mostly the number of legs that matters and the length of those legs is relatively unimportant. As we saw for the model of Ferrer i Cancho and Solé, the best networks in this case are star-like hub-and-spoke networks, and this is what we see in the present model too.

Thus this model interpolates between road-like and airline-like networks as the parameters are varied from one extreme to the other. Note that the parameter λ governing the cost of building or maintaining the network is held constant in Fig. 14.11. In principle, we could vary this parameter too, which would affect the total number of edges in the network. For higher λ sparser networks with fewer edges would be favored, while for lower λ we would see denser networks.

The work of Gastner and Newman still suffers from the drawback that the results are numerical only. More recently, however, some results for the model have been derived analytically by Aldous [15]. The interested reader is encouraged to consult his paper.

PROBLEMS

14.1 Consider the growing network model of Price, as described in Section 14.1.

a. From the results given in this chapter write down an expression in terms of the parameters a and c for the expected in-degree of the ith vertex added to the network just before the jth vertex is added, where $i < j$.
b. Hence show that the average probability of a directed edge from j to i in a network with n vertices, where $n \geq j$, is

$$p_{ij} = \frac{ca}{c+a} i^{-c/(c+a)} (j-1)^{-a/(c+a)}.$$

14.2 Consider Price's model as a model of a citation network, applied to publications in a single field, a field that is currently, say, ten years old.

a. Suppose that you are the author of the tenth paper published in the field. How long will it be from now before the expected number of citations your paper has within the field is equal to the expected number that the first paper published currently has?
b. Derive an expression for the average number of citations per paper to papers published between times τ_1 and τ_2, where time is defined as in Eq. (14.44).
c. Reasonable values of the model parameters for real citation networks are $c = 20$ and $a = 5$. For these parameter choices, what is the average number of citations to a paper in the first 10% of those published? And what is the average number for a paper in the last 10%?

These perhaps surprising numbers are examples of the first-mover advantage discussed in Section 14.3.1—the substantial bias of citation numbers in favor of the first papers published in a field.

14.3 Consider a model of a growing directed network similar to Price's model described in Section 14.1, but without preferential attachment. That is, vertices are added one by one to the growing network and each has c outgoing edges, but those edges now attach to existing vertices uniformly at random, without regard for degrees or any other vertex properties.

a. Derive master equations, the equivalent of Eqs. (14.7) and (14.8), that govern the distribution of in-degrees q in the limit of large network size.
b. Hence show that in the limit of large size the in-degrees have an exponential distribution $p_q = Ce^{-\lambda q}$, where C is a normalization constant and $\lambda = \ln(1 + 1/c)$.

14.4 Consider a model network similar to the model of Barabási and Albert described in Section 14.2, in which undirected edges are added between vertices according to a preferential attachment rule, but suppose now that the network does not grow—it starts off with a given number n of vertices and neither gains nor loses any vertices thereafter. In this model, starting with an initial

network of n vertices and some specified arrangement of edges, we add at each step one undirected edge between two vertices, both of which are chosen at random in direct proportion to degree k. Let $p_k(m)$ be the fraction of vertices with degree k when the network has m edges in total.

a. Show that when the network has m edges, the probability that vertex i will get a new edge upon the addition of the next edge is k_i/m.
b. Write down a master equation giving $p_k(m+1)$ in terms of $p_{k-1}(m)$ and $p_k(m)$. Be sure to give the equation for the special case of $k = 0$ also.
c. Eliminate m from the master equation in favor of the mean degree $c = 2m/n$ and take the limit $n \to \infty$ with c held constant to show that $p_k(c)$ satisfies the differential equation

$$c\frac{dp_k}{dc} = (k-1)p_{k-1} - kp_k.$$

d. Define a generating function $g(c,z) = \sum_{k=0}^{\infty} p_k(c)\,z^k$ and show that it satisfies the partial differential equation

$$c\frac{\partial g}{\partial c} + z(1-z)\frac{\partial g}{\partial z} = 0.$$

e. Show that $g(c, z) = f(c - c/z)$ is a solution of this differential equation, where $f(x)$ is any differentiable function of x.
f. The particular choice of f depends on the initial conditions on the network. Suppose the network starts off in a state where every vertex has degree one, which means $c = 1$ and $g(1, z) = z$. Find the function f that corresponds to this initial condition and hence find $g(c, z)$ for all values of c and z.
g. Show that, for this solution, the degree distribution as a function of c takes the form

$$p_k(c) = \frac{(c-1)^{k-1}}{c^k},$$

except for $k = 0$, for which $p_0(c) = 0$ for all c.

Note that this distribution decays exponentially in k, implying that preferential attachment does not, in general, generate a power-law degree distribution if the network is not also growing.

14.5 Consider a model of a growing network similar to Price's model described in Section 14.1, but in which the parameter a, which governs the rate at which vertices receive new incoming links when their current in-degree is zero, varies from vertex to vertex. That is the probability of a new edge attaching to vertex i is proportional to $q_i + a_i$, where q_i is the current in-degree and a_i is a specified parameter. In the context of citation networks, for example, a_i could be considered a measure of the intrinsic merit of a paper, controlling as it does the rate at which the paper gets citations immediately after first publication, when $q_i = 0$. (This differs from the model discussed in Section 14.4.4, where the preferential attachment term was multiplied by a varying factor to represent variations in the merit or fitness of vertices.)

a. Suppose that a_i is drawn at random from some stationary distribution with a welldefined mean. Show that, in the limit of large n, the probability that the $(n+1)$th vertex added to

the network attaches to a previous vertex i with in-degree q_i is $c(q_i + a_i)/n(c + \bar{a})$, where $\bar{a}$ is the average value of a_i.

b. Hence show that the in-degree distribution of the network satisfies the same master equations, (14.7) and (14.8), as Price's model, but with a replaced by ā.

(It immediately follows that the degree distribution of the network is also the same as for Price's model with the same substitution.)

14.6 Consider the following simple model of a growing network. Vertices are added to a network at a rate of one per unit time. Edges are added at a mean rate of β per unit time, where β can be anywhere between zero and ∞. (That is, in any small interval δt of time, the probability of an edge being added is $\beta\,\delta t$.) Edges are placed uniformly at random between any pair of vertices that exist at that time. They are never moved after they are first placed.

We are interested in the component structure of this model, which we will tackle using a master equation method. Let $a_k(n)$ be the fraction of vertices that belong to components of size k when there are n vertices in the network. Equivalently, if we choose a vertex at random from the n vertices currently in the network, $a_k(n)$ is the probability the vertex will belong to a component of size k.

a. What is the probability that a newly appearing edge will fall between a component of size r and another of size s? (You can assume that n is large and the probability of both ends of an edge falling in the same component is small.) Hence, what is the probability that a newly appearing edge will join together two pre-existing components to form a new one of size k?

b. What is the probability that a newly appearing edge joins a component of size k to a component of *any* other size, thereby creating a new component of size larger than k?

c. Thus write down a master equation that gives the fraction of vertices $a_k(n + 1)$ in components of size k when there are $n + 1$ vertices in total.

d. The only exception to the previous result is that components of size 1 appear at a rate of one per unit time. Write a separate master equation for $a_1(n + 1)$.

e. If a steady-state solution exists for the component size distribution, show that it satisfies the equations

$$(1+2\beta)a_1 = 1, \qquad (1+2\beta k)a_k = \beta k \sum_{j=1}^{k-1} a_j a_{k-j}.$$

f. Multiply by z^k and sum over k from 1 to ∞ and hence show that the generating function $g(z) = \sum_k a_k z^k$ satisfies the ordinary differential equation

$$2\beta \frac{\mathrm{d}g}{\mathrm{d}z} = \frac{1 - g/z}{1 - g}.$$

Unfortunately, the solution to this equation is not known, so for the moment at least we do not have a complete solution for the component sizes in the model.